

COURSE NOTES

GROUP THEORY

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February 26, 2026

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Chapter 1

Week 1

Part 1: Review and study basic concepts and basic theory by studying important classes of groups.

Part 2: Group actions. Permutation groups—O’Nan–Scott theorem.

Part 3: Linear groups—classical groups and subgroups.

Cyclic Groups.

Cyclic groups (denoted by C_n):

$$G = \langle g \rangle = \{1, g, g^2, g^3, \dots\}$$

(or $\{0, 1, 2, \dots\}$ sometimes written additively since commutative).

- Cyclic groups are uniquely determined by their order.
- Subgroups are cyclic.
- Factor groups are cyclic.

Proposition 1.0.1. $\text{Aut}(C_n) \simeq (\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{x} \mid (x, n) = 1\} = U(n)$.

Proof. For all $\theta \in \text{Aut}(C_n)$, $\theta(g) = g^i$ for some $i \in \mathbb{Z}$ and the integer i uniquely determines θ . Denote this automorphism by θ_i . We have $(i, n) = 1$ since g^i must have the same order as g . On the other hand, for all a such that $(a, n) = 1$, the map $g \mapsto g^a$ is an automorphism of C_n . Hence we have a surjective map

$$\psi: \text{Aut}(C_n) \rightarrow (\mathbb{Z}/n\mathbb{Z})^\times, \quad \theta_a \mapsto a \pmod{n}.$$

ψ is obviously a homomorphism and injective. □

- $\text{Aut}(C_p) = C_{p-1} = \langle \theta \rangle$ (use the fact that $\mathbb{F}_p^\times$ is cyclic).

Recall: let $F = GF(p^l)$.

- $F^+ = C_p^l = C_p \oplus \dots \oplus C_p$
- $F^\times = C_{p^l-1}$

Proposition 1.0.2. $\text{Aut}(C_{p^e}) = C_{p^{e-1}(p-1)}$ where $e \geq 2$ and p is an odd prime.

Proof. Let $r = \lambda$ or $p + \lambda$ where λ is a primitive root for C_p (so $\text{ord}_p(\lambda) = p - 1$). Since $(p + \lambda, p) = 1$, we have $(p + \lambda, p^e) = 1$, and

$$(p + \lambda)^{p^{e-1}(p-1)} \equiv 1 \pmod{p^e}.$$

It suffices to show $p^{e-1}(p-1)$ is the smallest integer n such that $(p + \lambda)^n \equiv 1 \pmod{p^e}$, or equivalently $\lambda^n \equiv 1 \pmod{p^e}$.

First note that $p - 1 \mid \text{ord}_{p^e}(r)$ since $r^{|r|} \equiv 1 \pmod{p}$ yields $r^{|r|} \equiv 1 \pmod{p}$, so $p - 1 \mid |r|$. Then it suffices to show $r^{p^{e-2}(p-1)} \not\equiv 1 \pmod{p^e}$.

When $e = 2$:

$$(p + \lambda)^{p-1} \equiv \lambda^{p-1} + p(p-1)\lambda^{p-2} \pmod{p^2}.$$

Here we use a lemma: for all λ primitive root of C_p , either λ or $p + \lambda$ is a primitive root for $(\mathbb{Z}/p^2\mathbb{Z})^\times$.

By above, $(p - 1) \mid |\lambda|$, so $|\lambda| = p - 1$ or $p(p - 1)$. If $|\lambda| = p - 1$, then

$$(p + \lambda)^{p-1} \equiv \lambda^{p-1} + p(p-1)\lambda^{p-2} \equiv p(p-1)\lambda^{p-2} \pmod{p^2}.$$

Since $p \nmid \lambda^{p-2}$, we have $(p + \lambda)^{p-1} \not\equiv 1 \pmod{p^2}$, so $p + \lambda$ is a primitive root for $(\mathbb{Z}/p^2\mathbb{Z})^\times$.

Suppose it works for $e \leq n$, i.e., $r^{p^{n-2}(p-1)} \not\equiv 1 \pmod{p^n}$. Since $r^{p^{n-2}(p-1)} \equiv 1 \pmod{p^{n-1}}$, we have

$$r^{p^{n-2}(p-1)} = 1 + dp^{n-1}$$

for some $(d, p) = 1$. Then

$$r^{p^{n-1}(p-1)} = (1 + dp^{n-1})^p \equiv 1 + dp^n + \binom{p}{2} d^2 p^{2n-2} \equiv 1 + dp^n \not\equiv 1 \pmod{p^{n+1}}.$$

□

Remark 1.0.3. We actually proved that if r is a primitive root for $(\mathbb{Z}/p^2\mathbb{Z})^\times$, then r is a primitive root for $(\mathbb{Z}/p^k\mathbb{Z})^\times$ for any k . And we prove existence of such r by noting that if λ is a primitive root for C_p , then λ or $p + \lambda$ is a primitive root of $(\mathbb{Z}/p^2\mathbb{Z})^\times$.

Computation: Prove that each Sylow q -subgroup is cyclic for each prime q , and each finite abelian group is a direct sum of cyclic groups.

(1) Sylow p -subgroup is cyclic.

Claim: $1 + p$ is an element of order $p^{\alpha-1}$ in $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$. (Show that $(1 + p)^{p^{\alpha-1}} \equiv 1 \pmod{p^\alpha}$ but $(1 + p)^{p^{\alpha-2}} \not\equiv 1 \pmod{p^\alpha}$.)

(2) Sylow q -subgroup is cyclic, $q \neq p$.

The map

$$\mathbb{Z}/p^\alpha\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}, \quad a + (p^\alpha) \mapsto a + (p)$$

is a ring homomorphism which gives a surjective homomorphism from $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ onto $(\mathbb{Z}/p\mathbb{Z})^\times$. The latter group is cyclic of order $p - 1$. The kernel of this map is of order $p^{\alpha-1}$, hence for all primes $q \neq p$, the Sylow q -subgroup of $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ maps isomorphically into the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$. All Sylow subgroups of $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ are therefore cyclic, so $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is cyclic.

Proposition 1.0.4. $\text{Aut}(C_{2^e}) = C_{2^{e-2}} \times C_2$ for $e \geq 2$.

Note that C_2 and $\mathbb{Z}/4\mathbb{Z}$ have primitive roots (1 and 3 respectively), but not for $\mathbb{Z}/2^\alpha\mathbb{Z}$ when $\alpha \geq 3$.

(1) If $a = 2b + 1$ for some b , $k \geq 3$, then $a^{2^{k-2}} \equiv 1 \pmod{2^k}$.

When $k = 3$: $a^2 = (2b+1)^2 = 4b^2 + 4b + 1 = 1 + 4b(b+1) \pmod{2^3}$. Since $8 \mid 4b(b+1)$, we have $a^2 \equiv 1 \pmod{2^3}$.

Suppose it holds for $k \leq n$. Then $(1 + \ell \cdot 2^k)^2 = \ell^2 \cdot 2^{2k} + \ell \cdot 2^{k+1} + 1 \equiv 1 \pmod{2^{k+1}}$.

(2) 5 is an element of order 2^{k-2} in $(\mathbb{Z}/2^k\mathbb{Z})^\times$. (Show that $(1 + 2^2)^{2^{k-2}} \equiv 1 \pmod{2^k}$ and $(1 + 2^2)^{2^{k-3}} \equiv 1 + 2^{k-1} \not\equiv 1 \pmod{2^k}$.)

(3) Since $(\mathbb{Z}/2^k\mathbb{Z})^\times$ is finite abelian, it is a direct product of cyclic groups. It has an element of order 2^{k-2} , then $(\mathbb{Z}/2^k\mathbb{Z})^\times \simeq C_{2^{k-2}} \times C_2$.

Proposition 1.0.5.

$$\text{Aut}(C_n) \cong \text{Aut}(C_{p_1^{e_1}}) \times \cdots \times \text{Aut}(C_{p_t^{e_t}}), \quad n = p_1^{e_1} \cdots p_t^{e_t}.$$

defined by

$$r \mapsto (r + A_1, r + A_2, \dots, r + A_k)$$

is a ring homomorphism with kernel $A_1 \cap A_2 \cap \cdots \cap A_k$.

If for each $i, j \in \{1, 2, \dots, k\}$ with $i \neq j$ the ideals A_i and A_j are coprime ($A_i + A_j = R$), then this map is surjective and $A_1 \cap A_2 \cap \cdots \cap A_k = A_1 A_2 \cdots A_k$, so

$$R/(A_1 A_2 \cdots A_k) = R/(A_1 \cap A_2 \cap \cdots \cap A_k) \simeq \prod_{i=1}^k R/A_i.$$

Theorem 1.0.6 (Chinese Remainder Theorem). *Let $m_1, \dots, m_n$ be pairwise coprime integers. Then*

$$\mathbb{Z}/(m_1 \cdots m_n)\mathbb{Z} \simeq (\mathbb{Z}/m_1\mathbb{Z}) \times \cdots \times (\mathbb{Z}/m_n\mathbb{Z}).$$

Proof sketch of general version:

1. The homomorphism with $\ker \varphi = A_1 \cap \cdots \cap A_n$ is obvious.
2. Surjective if coprime.

Lemma 1.0.7. *Let R be a ring with identity. If I_1, I_2, J are ideals of R such that I_1 and J are coprime, and I_2 and J are coprime, then $I_1 I_2$ and J are coprime.*

Proof. There exist $a_1 \in I_1, b_1 \in J$ such that $a_1 + b_1 = 1$, and $a_2 \in I_2, b_2 \in J$ such that $a_2 + b_2 = 1$. Then

$$(a_1 + b_1)(a_2 + b_2) = 1 = \underbrace{a_1 a_2}_{\in I_1 I_2} + \underbrace{a_1 b_2 + b_1 a_2 + b_1 b_2}_{\in J}.$$

□

By the lemma, I_1 and $I_2 \cdots I_n$ are coprime, so there exist $a \in I_1, b \in I_2 \cdots I_n$ such that $a + b = 1$. Let $x_1 = 1 - a = b$. Then $x_1 + I_1 = 1 - a + I_1 = 1 + I_1$ and $x_1 + I_i = b + I_i = I_i$ for $i \neq 1$. Hence $\varphi(x_1) = (1 + I_1, I_2, \dots, I_n)$. Similarly we find x_i for each i , so for all $x = (r_1 + I_1, \dots, r_n + I_n) \in \prod_{i=1}^n R/A_i$, there exists $r = r_1 x_1 + \cdots + r_n x_n$ such that $x = \varphi(r)$.

3. If coprime, then $A_1 \cap \cdots \cap A_n = A_1 \cdots A_n$.

This is obvious for $n = 1$. By induction, it suffices to prove $A_1 \cap A_2 \subseteq A_1 A_2$. Indeed, if $a \in A_1 \cap A_2$ and $x \in A_1, y \in A_2$ satisfy $x + y = 1$, then $a = a(x + y) = ax + ay \in A_1 A_2$ since R is commutative.

Since the isomorphism in the Chinese Remainder Theorem is an isomorphism of rings, in particular the groups of units on both sides must be isomorphic. Thus

$$(\mathbb{Z}/mn\mathbb{Z})^\times \simeq (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$$

whenever $(m, n) = 1$.

Remark 1.0.8. *There are 3 elements of order 2 in $\text{Aut}(C_{2^e})$:*

- $g \mapsto g^{-1}$
- $g \mapsto g^{2^{e-1}+1}$, e.g., using $g \mapsto g^5$
- $g \mapsto g^{2^{e-1}-1}$ (which equals $(g^{2^{e-1}+1})^{-1}$)

Dihedral Group $G = D_{2n}$

The dihedral group is $G = \langle a, b \rangle$ with $a^2 = b^2 = 1$.

Example 1.0.9. *Let*

$$a = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 & 2 \\ \frac{1}{2} & 0 \end{pmatrix} \in GL(2, \mathbb{R}).$$

Let $|a| = |b| = 2$ with ab having infinite order. Then $G = \langle a, b \rangle$ is an infinite dihedral group.

Remark 1.0.10. $|G|$ is finite if and only if $|ab|$ is finite.

The elements of G are $a, ab, aba, abab, \dots$ and $b, ba, bab, baba, \dots$. As soon as we know the first term in these lists to give the identity element, we know G . It can't be elements like $ababa$, since $1 = ababa = bab = a$. So it must be $(ab)^m = 1$ for some m , and then G is finite. Note that $(ab)^m = 1 \Leftrightarrow (ba)^m = 1$.

$$G = \{(ab)^i, (ba)^j, (ab)^i a, (ba)^j b\}$$

$$\Rightarrow G = \{(ab)^i, b(ab)^i \mid i \in \mathbb{Z}\}.$$

Since $(ba)^j = (ba)^j b^2 = b(ab)^j b = ((ab)^j)^{-1}$ and $(ab)^i a = b(ba)^{i+1} = b((ab)^i)^{-1}$.

If G is finite, $|ab| = n$, so $|G| = 2n$ and $|ab| = \frac{|G|}{2}$. Thus $G = \langle ab, b \rangle = \langle ab, a \rangle = \langle ab \rangle : \langle b \rangle = C_n : C_2$.

Definition 1.0.11. Let G be a group, $A, B < G$. Then G is called a semidirect product of A by B , denoted $A : B$ or $A \rtimes B$, if:

1. $A \triangleleft G$
2. $A \cap B = 1$
3. $G = AB$

Example 1.0.12. Consider $\text{Aut}(D_{2n})$. Let $g \mapsto g^i$ where $(i, n) = 1$. If n is odd, then $b \mapsto g^j b$ for some j gives an automorphism of D_{2n} .

Then $\text{Aut}(D_{2n}) \simeq C_n : \text{Aut}(C_n)$.

In particular:

- $\text{Aut}(D_{2p}) = C_p : C_{p-1}$
- $\text{Aut}(D_{2p^l}) = C_{p^l} : C_{p^{l-1}(p-1)}$
- $|\text{Aut}(D_{2n})| = n\phi(n)$

Remark 1.0.13. Hint: $G = D_{2n} = \langle g \rangle : \langle b \rangle = C_n : C_2$. If $\sigma \in \text{Aut}(G)$, then $g^\sigma = g^i$ since $\langle g^\sigma \rangle$ has odd order.

Conjugacy classes of D_{2n}

Let $D_{2n} = \langle g \rangle : \langle b \rangle = \{g^i, g^j b \mid 0 \leq i, j \leq n-1\}$.

Case 1: n is odd.

1. Elements of odd order form conjugacy classes $\{g^i, g^{-i}\}$.
2. Elements of even order are all conjugate.

Proof. Claim: $g^j b$ and $g^{j+1} b$ are conjugate.

Observe that:

1. D_{2n} has exactly n elements of order 2: $g^j b$ for $0 \leq j \leq n-1$.
2. $g^{-i} b g^i$ with $0 \leq i \leq n-1$ are of order 2 and distinct, and are obviously conjugate.
Thus $g^i b = g^{-j} b g^j$ for some j , and all $g^i b$ are conjugate.

□

Case 2: n is even.

Chapter 2

Week 2

Constructing groups and finding $\text{Aut}(G)$ for the new G .

Construction.

Let G be a group and let $H \leq \text{Aut}(G)$. Let $X = \{gh \mid g \in G, h \in H\}$ such that $(g_1 h_1)(g_2 h_2) = (g_1 g_2^{h_1^{-1}})(h_1 h_2)$. Then X is a group denoted by $G \cdot H$ in which $G \triangleleft X$ and $X/G \simeq H$.

Example 2.0.1 (Cyclic group). Let $G = Z_7 = \langle g \rangle$. Then $\text{Aut}(G) = Z_6$.

1. Let $\sigma \in \text{Aut}(G)$ of order 2 (so $g^\sigma = g^{-1}$).

Then $X = G \cdot \langle \sigma \rangle = \langle g \rangle \cdot \langle \sigma \rangle$ with $(g\sigma)(g\sigma) = gg^\sigma = 1$. This gives

$$X = \langle g, \sigma \mid g^\sigma = g^{-1}, |g| = 7, |\sigma| = 2 \rangle = Z_7 \cdot Z_2 = D_{14}.$$

2. Let $z \in \text{Aut}(G)$ of order 3.

Then $g^z = g^2$. Thus $X = \langle g, z \rangle$ where $|g| = 7$, $|z| = 3$, $g^z = g^{(2)}$, so $X \simeq Z_7 : Z_3$.

Example 2.0.2 (General cyclic group). Let $G = \langle g \rangle = Z_p$ where p is prime. Then $\text{Aut}(G) = Z_{p-1}$.

Let $\sigma \in \text{Aut}(G)$ such that $|\sigma| = l$. Then $l \mid (p-1)$ and $\langle \sigma \rangle = Z_l$. We have $X = G \cdot \langle \sigma \rangle = Z_p \cdot Z_l = \langle g, \sigma \rangle$ where $|g| = p$, $|\sigma| = l$, $g^\sigma = g^\lambda$ for some λ .

If $|\sigma| = 2$, then $X = D_{2p}$.

If $l = p-1$, then $X = \text{AGL}(1, p)$ and $Z(X) = 1$.

In general, for $G = Z_n$, we have $X = Z_n \cdot H$ where $H \leq U(n)$.

Example 2.0.3. Let $G = Z_{35} = Z_5 \times Z_7$. Then $H \leq \text{Aut}(G) = Z_4 \times Z_6 = \langle b_1 \rangle \times \langle b_2 \rangle$.

We have $X = Z_{35} : H < (Z_5 : Z_4) \times (Z_7 : Z_6)$.

For instance, $X_0 = Z_{35} : \langle b_1^2 b_2^3 \rangle = D_{70}$.

Let $G = D_{2n} = Z_n : Z_2$. Then $\text{Aut}(G) = Z_n : U(n) = Z_n : \text{Aut}(Z_n)$.

Example 2.0.4 (Finite field). Let $F = \text{GF}(p^f)$ and let $G = F^+ \simeq Z_p^f$, $H = F^\times \simeq Z_{p^f-1}$. Define a group $X = \{gh \mid g \in G, h \in H\}$ such that

$$(g_1 h_1)(g_2 h_2) = g_1 g_2^{h_1^{-1}} h_1 h_2$$

where $g_2^{h_1^{-1}} = g_2 h_1^{-1}$ in F under multiplication.

For $g, h \in F$, the element h induces an automorphism of $G = F^+$ by multiplication: $(a + b)^h = (a + b)h = ah + bh = a^h + b^h$. Thus

$$X = G : H = Z_p^f : Z_{p^f-1}$$

which is a Singer cycle.

Moreover,

$$\text{Aut}(X) = X : \langle \tau \rangle = (Z_p^f : Z_{p^f-1}) : Z_f = (Z_p^f : \langle g \rangle) : \langle \tau \rangle$$

where $\tau : g \mapsto g^p$ is the Frobenius automorphism.

In particular, $\text{Aut}(Z_p : Z_{p-1}) \simeq Z_p : Z_{p-1}$.

Example 2.0.5. Assume $p = 3$, $f = 2$. Then

$$\text{Aut}(X) = (Z_3^2 : Z_8) : Z_2$$

where $z : g \mapsto g^3$.

We have $\langle g \rangle$ and $\langle z \rangle$ with presentation

$$\langle a, b \mid a^4 = 1, a^2 = b^2, a^b = -a \rangle.$$

Let $a = g^2$ and $b = gz$. Let $\langle a, b \rangle = Q_8 = \{1, -1, i, -i, j, -j, k, -k\}$.

$$b^2 = gz \cdot gz = gg^{-1}z^2 = z^2 = g^4$$

and $|b| = 4$.

$$a^2 = b^2$$

$$a^b = (gz)^{-1}g^2(gz) = g^6 = (g^2)^{-1} = a^{-1}$$

$$b^a = b^{-1}$$

We have $\langle g \rangle \triangleleft \langle g^2, gz \rangle$.

Also $\text{Aut}(X) = (Z_3^2 : Z_8) : Z_2 = Z_3^2 : (Z_8 : Z_2) > Z_3^2 : Q_8$.

HW: All elements of $Z_3^2 : Q_8$ of order 3 are conjugate.

Definition 2.0.6 (Group Action). *Let G be a group and Ω a set. We say that G acts on Ω if:*

1. *Each element g of G is a one-to-one map from Ω to Ω . The image of $\omega \in \Omega$ under g is denoted by ω^g .*
2. *$\omega^1 = \omega$ for all $\omega \in \Omega$.*
3. *$(\omega^{g_1})^{g_2} = \omega^{g_1 g_2}$.*

Example 2.0.7. 1. $\text{GL}(\Omega)$ acts on $\Omega = F^n$.

2. *Conjugation:* $\Omega = G$, $\omega^g = g^{-1}\omega g$. The conjugacy class of g is $C(g) = \{g^x \mid x \in G\}$.

3. *Right multiplication:* $\Omega = G$, $\omega^g = \omega g$.

Lemma 2.0.8 (N/C -lemma). *For $H < G$, we have $C_G(H) \trianglelefteq N_G(H)$ and $N_G(H)/C_G(H) \lesssim \text{Aut}(H)$.*

In particular, if $H \trianglelefteq G$, then $G/C_G(H) \lesssim \text{Aut}(H)$.

Proof. $N_G(H)$ acts on H by conjugation with kernel equal to $C_G(H)$. □

Chapter 3

Week 3

Theorem 3.0.1 (Cauchy). *If $p \mid |G|$, then $\exists g \in G$ such that $|g| = p$.*

Let G act on a set Ω . Then G partitions Ω into orbits:

$$\Omega = \Omega_1 \dot{\cup} \Omega_2 \dot{\cup} \cdots \dot{\cup} \Omega_m$$

Theorem 3.0.2 (Orbit-Stabilizer Theorem).

$$|G| = |\Omega_i| \cdot |G_{w_i}| \quad \text{where } w_i \in \Omega_i \text{ and } G_{w_i} = \{g \in G \mid w_i^g = w_i\}$$

Example 3.0.3 (Conjugation action). *Assume $\Omega = G$ and G acts on Ω by conjugation. Then $G = \{1\} \cup C(x_1) \cup \cdots \cup C(x_t)$. Then $|G| = |C(x_i)| \cdot |C_G(x_i)|$. In particular, $|C(x_i)| \mid |G|$.*

Theorem 3.0.4. *If $|G| = p^e$ with p prime, then $Z(G) \neq 1$.*

Proof. $G = \{1\} \dot{\cup} C(x_1) \dot{\cup} \cdots \dot{\cup} C(x_t)$. Then

$$\begin{aligned} |G| &= p^e = 1 + |C(x_1)| + \cdots + |C(x_t)| \\ &= 1 + p^{e_1} + \cdots + p^{e_t} \quad \text{where } e_i \geq 1. \end{aligned}$$

It follows that $|C(x_i)| = 1$ for some i . So $1 \neq x_i \in Z(G)$. □

Remark 3.0.5. *If x and y are conjugate, then $|x| = |y|$. The converse is not necessarily correct, e.g., in $C_n = \langle g \rangle$, g and g^{-1} are not conjugate.*

Exercise 3.0.6. *If G is a group satisfying that elements of the same order are conjugate, then $G \cong S_2$ or S_3 . (Solved independently by W. Feit.)*

Sylow Theorems.

Let G be finite, $|G| = p^e m$ with p prime and $\gcd(p, m) = 1$.

Definition 3.0.7 (Sylow p -subgroup). *If G has a subgroup H such that $|H| = p^e$, then H is called a Sylow p -subgroup.*

Remark 3.0.8. *Existence of $H \Rightarrow$ Cauchy's theorem.*

Theorem 3.0.9 (First Sylow Theorem). *Sylow p -subgroups exist for all $p \mid |G|$.*

Proof. Use the conjugation equation:

$$G = \{1\} \dot{\cup} C(x_1) \dot{\cup} \cdots \dot{\cup} C(x_t)$$

$$|G| = 1 + |C(x_1)| + \cdots + |C(x_t)| = |Z(G)| + |C(x_1)| + \cdots + |C(x_s)|$$

(1) $p \nmid |Z(G)|$. Then $\exists C(x_i)$ such that $\gcd(|C(x_i)|, p) = 1$ and $C(x_i) \neq \{1\}$. By the orbit-stabilizer theorem, since $|G| = |C(x_i)| \cdot |C_G(x_i)|$, we have $p^e \mid |C_G(x_i)|$, i.e., $C_G(x_i)$ is a subgroup of G of order divisible by p^e with $|C_G(x_i)| < |G|$. By induction, $C_G(x_i)$ has a Sylow p -subgroup P . Then P is a Sylow p -subgroup of G .

(2) $p \mid |Z(G)|$. $Z(G)$ is an abelian group. We first prove Cauchy's theorem for abelian groups.

Let G be abelian. Pick $1 \neq x \in G$. If $p \mid |x|$, then $x^{|x|/p}$ is of order p . If $p \nmid |x|$, then $|G/\langle x \rangle|$ is divisible by p . By induction, $\exists a \in G$ such that $a\langle x \rangle$ is of order p . Then $p \mid |a|$, since $(a\langle x \rangle)^{|a|} = 1$. Then $a^{|a|/p}$ is of order p .

Thus $\exists x \in Z(G)$ of order p . So $p^{e-1} \mid |G/\langle x \rangle|$. By induction, $G/\langle x \rangle$ has a subgroup of the form $H/\langle x \rangle$ of order p^{e-1} . Then H is a subgroup of order p^e . $\square$

Theorem 3.0.10 (Second Sylow Theorem). *Sylow p -subgroups are conjugate.*

Proof. Let $\Omega = \{P_1, P_2, \dots, P_n\}$ be the set of Sylow p -subgroups. Let $P = P_1$ and let $\Delta = \{Px \mid x \in G\}$. Then $|\Delta| = [G : P]$. Then G acts on Δ by right multiplication: $g: Px \mapsto Pxg$.

Then $|\Delta| = \frac{|G|}{|P|}$, which is coprime to p , and the action of G on Δ is transitive.

Let $H < G$ such that $|H| = p^d \leq p^e$. Then H acting on Δ is intransitive (otherwise $|\Delta| \mid p^d$, contradiction). Suppose H acting on Δ partitions Δ into orbits $\Delta_1, \dots, \Delta_r$. Then $|\Delta_i| \mid |H|$ and $|\Delta| = |\Delta_1| + \cdots + |\Delta_r|$. So $|\Delta_i| = 1$ for some i since $p \nmid |\Delta|$, i.e., $Pxh = Px$ for some x , for each $h \in H$. $\square$

$$xHx^{-1} < P$$

$$H < x^{-1}Px$$

If $|H| = p^e$, then $H = x^{-1}Px$.

Theorem 3.0.11 (Third Sylow Theorem). *Let $|G| = p^e m$ with $\gcd(p, m) = 1$ and let n_p denote the number of Sylow p -subgroups of G . Then $n_p \mid m$ and $n_p \equiv 1 \pmod{p}$.*

Group Actions.

Let G be finite. Let $N \trianglelefteq G$ be minimal. Then $\bar{G} = G/N$ is a group with $|\bar{G}| < |G|$. So $G = N \cdot \bar{G}$ (extension).

Lemma 3.0.12. *If N is a minimal normal subgroup of G , then $N = S_1 \times \cdots \times S_r$ where $S_1 \simeq S_2 \simeq \cdots \simeq S_r$ are simple. i.e., $N \simeq S^r = S_1 \times \cdots \times S_r$. Furthermore, $N \simeq C_p^r$ or S is nonabelian.*

Proof. If N is simple, then we are done.

Assume that N is not simple. Let M be a minimal normal subgroup of N . By induction, $M = T_1 \times \cdots \times T_m$ where $T_1 \simeq T_2 \simeq \cdots \simeq T_m$ are simple.

For $g \in G$, the conjugate $M^g = g^{-1}Mg$ is a minimal normal subgroup of N . By minimality, either $M^g = M$ or $M \cap M^g = \{1\}$.

The normal closure of M under G is

$$M^G = \langle M^g \mid g \in G \rangle = M \times M^{g_1} \times \cdots \times M^{g_l}.$$

Note that if $M \cap M^g = 1$, then $\langle M, M^g \rangle = M \times M^g$ since $M \trianglelefteq N$ and $M^g \trianglelefteq N$.

Also, since $M = T_1 \times \cdots \times T_m$, we have $M^g = T_1^g \times \cdots \times T_m^g$. Hence $M^g \simeq T^m$.

Since $M^G \trianglelefteq G$ and $M^G \leq N$, we have $N = M^G = T^{ml}$.

Now change notation: $M \trianglelefteq G$ minimal. Then

$$M = \begin{cases} C_p^r \\ T \times \cdots \times T \text{ (nonabelian)} \end{cases}$$

We have

$$G/C_G(M) \cong N_G(M)/C_G(M) \leq \text{Aut}(M) \quad \text{where } M \trianglelefteq G.$$

Question: For T_i nonabelian, we want to find $\text{Aut}(T \times \cdots \times T_r)$.

For $g \in \text{Aut}(M)$, we have $T_i^g \trianglelefteq M$ since $T_i \trianglelefteq M$.

Claim: $T_i^g = T_j$ for some j .

Proof. We have $T_i^g \trianglelefteq T_1 \times \cdots \times T_r = M$.

Suppose $T_i^g \neq T_j$ for all j . Let $L = T_2 \times \cdots \times T_r$. Then $T_i^g \cap L = 1$ since $T_i^g \cap L \trianglelefteq T_i^g$ and T_i^g is simple.

Since $T_i^g \subseteq M = T_1 \times \cdots \times T_r$, there exists $(t_1, y) \in T_i^g$ where $t_1 \neq 1$ and $y \neq 1, y \in L$.

For all $t \in T_1$:

$$(t_1, y)^{(t,1)} = (t_1^t, y).$$

Since T_i is nonabelian and simple, $Z(T_i) \trianglelefteq T_i$ implies $Z(T_i) = 1$. Hence there exists $t \in T_1$ such that $t_1^t \neq t_1$, and

$$(t_1, y)^{-1}(t_1^t, y) = (t_1^{-1}t_1^t, 1) = (t_0, 1) \in T_i^g.$$

Then

$$\langle (t_0, 1)^{(t,1)} \mid (t, 1) \in T_1 \times L \rangle \trianglelefteq T_i^g.$$

$\langle (t_0^t, 1) \mid t \in \Pi \rangle \subseteq T_1 \Rightarrow \langle \rangle = T_1 \Rightarrow \pi_1^g = T_1$.

On the other hand, $T_i^g < T_2 \times \cdots \times T_r$, continue.

$\Rightarrow \text{Aut}(M)$ induces a permutation group on $\{T_1, T_2, \dots, T_r\}$ with kernel $\text{Aut}(T_1) \times \cdots \times \text{Aut}(T_r) =: K$.

s.t. $\text{Aut}(M)/K \cong S_r \Rightarrow \text{Aut}(M) = K \cdot S_r = \text{Aut}(T)^r \cdot S_r$.

Example 3.0.13. *Structure of G where $|G| = 72$.*

Suppose it is solvable, $\exists N \trianglelefteq_{\min} G$.

$$N \simeq T^r, \quad |T^r| \mid 72 \Rightarrow N \cong C_3^2 \text{ or } C_2^3.$$

By conjugation action, $T \trianglelefteq \text{Aut}(T)$, and $\text{Aut}(T)/T$ is relatively small.

Chapter 4

Week 4

(Burnside) If G is a finite group whose Sylow subgroups are all cyclic, then G is solvable.

Definition 4.0.1 (Transfer). Let G be a finite group and $H < G$. Let $\Omega = [G : H] = \{Hx \mid x \in G\} = \{Hx_1, \dots, Hx_n\}$. For $g \in G$, define

$$\hat{g}: Hx_i \rightarrow Hx_j = Hx_i g.$$

So $x_i g \in Hx_j$ and $x_i g = h_i(g)x_j$ for some $h_i(g) \in H$. The map $\hat{g}$ induces a permutation on $\{1, 2, \dots, n\}$, denoted by $\pi(g)$, i.e., $i^{\pi(g)} = j$. Let

$$h_i(g) = x_i g x_{\pi(g)(i)}^{-1} \in H.$$

The transfer from G to H/H' is defined by

$$V_{G \rightarrow H}: G \rightarrow H/H', \quad g \mapsto \prod_{i=1}^n h_i(g) H' = \prod_{i=1}^n x_i g x_{\pi(g)(i)}^{-1} H'.$$

Lemma 4.0.2. 1. $V_{G \rightarrow H}$ is a homomorphism from G to H/H' .

2. $V_{G \rightarrow H}$ does not depend on the choice of coset representatives $x_1, \dots, x_n$.

3. Writing $\hat{g}$ in cycle form,

$$V_{G \rightarrow H}(g) = \prod_{i=1}^t \left(x_i g^{\ell_i} x_i^{-1} \right) H'.$$

Proof. (1) For $g_1, g_2 \in G$,

$$\begin{aligned}
V_{G \rightarrow H}(g_1 g_2) &= \prod_{i=1}^n h_i(g_1 g_2) H' \\
&= \prod_{i=1}^n x_i g_1 g_2 x_{\pi(g_1 g_2)(i)}^{-1} H' \\
&= \prod_{i=1}^n x_i g_1 x_{\pi(g_1)(i)}^{-1} \cdot x_{\pi(g_1)(i)} g_2 x_{\pi(g_1 g_2)(i)}^{-1} H' \\
&= \prod_{i=1}^n x_i g_1 x_{\pi(g_1)(i)}^{-1} H' \cdot \prod_{i=1}^n x_{\pi(g_1)(i)} g_2 x_{\pi(g_1 g_2)(i)}^{-1} H' \\
&= V_{G \rightarrow H}(g_1) V_{G \rightarrow H}(g_2),
\end{aligned}$$

where we used that H/H' is abelian.

(2) Let $y_1, \dots, y_n \in G$ such that $H y_i = H x_i$ for $1 \leq i \leq n$. Then $y_i = t_i x_i$ for some $t_i \in H$. For $g \in G$, let $y_i g = k_i(g) y_{\pi(g)(i)}$. Then $k_i(g) = y_i g y_{\pi(g)(i)}^{-1} \in H$. The corresponding transfer is

$$\begin{aligned}
\tilde{V}_{G \rightarrow H}(g) &= \prod_{i=1}^n k_i(g) H' = \prod_{i=1}^n y_i g y_{\pi(g)(i)}^{-1} H' \\
&= \prod_{i=1}^n t_i x_i g (t_{\pi(g)(i)} x_{\pi(g)(i)})^{-1} H' \\
&= \prod_{i=1}^n t_i (x_i g x_{\pi(g)(i)}^{-1}) t_{\pi(g)(i)}^{-1} H'.
\end{aligned}$$

Since H/H' is abelian,

$$\tilde{V}_{G \rightarrow H}(g) = \prod_{i=1}^n x_i g x_{\pi(g)(i)}^{-1} H' \cdot \prod_{i=1}^n t_i H' \cdot \prod_{i=1}^n t_{\pi(g)(i)}^{-1} H' = \prod_{i=1}^n x_i g x_{\pi(g)(i)}^{-1} H' = V_{G \rightarrow H}(g).$$

(3) The permutation $\hat{g}$ on $\Omega = \{Hx_1, \dots, Hx_n\}$ partitions Ω into t orbits:

$$(Hx_1, Hx_1 g, \dots, Hx_1 g^{\ell_1-1}), \quad (Hx_2, Hx_2 g, \dots, Hx_2 g^{\ell_2-1}), \quad \dots, \quad (Hx_t, \dots, Hx_t g^{\ell_t-1}),$$

where $Hx_i g^{\ell_i} = Hx_i$. Take

$$x_1, x_1 g, \dots, x_1 g^{\ell_1-1}, \quad x_2, x_2 g, \dots, x_2 g^{\ell_2-1}, \quad \dots, \quad x_t, x_t g, \dots, x_t g^{\ell_t-1}$$

as coset representatives. The corresponding transfer is

$$V_{G \rightarrow H}(g) = \prod_{i=1}^t (x_i g^{\ell_i} x_i^{-1}) H'.$$

□

$$\begin{aligned} V_{G \rightarrow H}(g) &= \prod x_i g (x_i g)^{-1} (x_i g) g (x_i g)^{-1} \cdots x_i g^{l_i-1} g x_i^{-1} H \\ &= \prod (x_i g^{l_i} x_i^{-1}) H' \end{aligned}$$

Definition 4.0.3 (*p*-nilpotent). A group G is called *p*-nilpotent if $G = NP$ where $N \triangleleft G$ and P is a Sylow *p*-subgroup and $N \cap P = 1$ (equivalently, if G has a normal subgroup N such that $|G/N| = |G|_p = |P|$, where p is a prime divisor of $|G|$ and P is any Sylow *p*-subgroup of G).

Theorem 4.0.4 (Burnside). Let G be a finite group and P a Sylow *p*-subgroup. If $N_G(P) = C_G(P)$, then G is *p*-nilpotent.

Proof. (Use transfer to construct $\ker V_{G \rightarrow P} = N$.) Since $N_G(P) = C_G(P)$, P is abelian ($P \leq N_G(P)$ hence $P \subseteq C_G(P)$), so $P' = 1$.

Let $g \in P \setminus \{1\}$. Then

$$V_{G \rightarrow P}(g) = \prod_{i=1}^t x_i g^{l_i} x_i^{-1}.$$

As $g^{l_i}, x_i g^{l_i} x_i^{-1} \in P$ and P is abelian, $C_G(g^{l_i}) \geq P$. Also $C_G(x_i g^{l_i} x_i^{-1}) = x_i C_G(g^{l_i}) x_i^{-1} \geq P$. Hence $C_G(g^{l_i}) \geq P$ and $C_G(g^{l_i}) \geq x_i^{-1} P x_i$.

By Sylow's theorem, there exists $u \in C_G(g^{l_i})$ such that

$$u^{-1}(x_i^{-1} P x_i)u = P,$$

i.e., $(x_i u)^{-1} P (x_i u) = P$.

Thus $x_i u \in N_G(P) = C_G(P) \Rightarrow x_i \in C_G(P) \leq C_G(g^{l_i})$.

Therefore

$$V_{G \rightarrow P}(g) = \prod_{i=1}^t g^{l_i} = g^{l_1 + \cdots + l_t} = g^{|G:P|} \neq 1$$

(since $g \in P \setminus \{1\}$), which yields $V_{G \rightarrow P}(P) = P$ (since it is injective by above discussion and $|G| < \infty$) $\Rightarrow V_{G \rightarrow P}(G) = P$.

Let $K = \ker V_{G \rightarrow P} \triangleleft G$. Then by the isomorphism theorem, $G/K \simeq P$, so $K \triangleleft G$. $\square$

Theorem 4.0.5 (Burnside). Let G be a finite group of which Sylow subgroups are all cyclic. Let p be the smallest prime divisor of $|G|$ and let P be a Sylow *p*-subgroup of G . Then:

1. G is *p*-nilpotent;
2. G is solvable.

Proof. Now P is cyclic and $N_G(P)/C_G(P) \lesssim \text{Aut}(P)$ where $|\text{Aut}(P)| = p^{e-1}(p-1)$, whose prime divisors are $\leq p$. Then we can only have $N_G(P)/C_G(P) = 1$ since $p^e \mid |N_G(P)|$ as $P \leq C_G(P)$, and all prime divisors of the LHS are $> p$. Then $N_G(P) = C_G(P)$ and so G has a normal subgroup K with $|K| = |G|/|P|$.

Let p_1 be the smallest prime divisor of $|K|$. Then K is p_1 -nilpotent and has a normal subgroup K_1 with $|K_1| = |K|/|P_1|$. Repeating this process, we have that

$$G \triangleright K \triangleright K_1 \triangleright \cdots \triangleright Q$$

where Q is a Sylow q -subgroup, where q is the largest prime divisor of $|G|$. $\square$

Remark 4.0.6. $Q \triangleleft G$.

Proof. Suppose $Q \triangleleft K_n \triangleleft K_{n-1}$. Then $\forall g \in K_{n-1}$, $g^{-1}Qg \subseteq g^{-1}K_n g = K_n$. Since $Q \triangleleft K_n$, K_n has only one Sylow q -subgroup, then $g^{-1}Qg = Q$, hence $Q \triangleleft K_{n-1}$. Repeating this process, $Q \triangleleft G$. $\square$

Corollary 4.0.7. *If a Sylow 2-subgroup of G is cyclic, then G is 2-nilpotent hence solvable.*

Proof. Need to use Feit–Thompson: $|G|$ odd $\Rightarrow G$ solvable. Then N and G/N both solvable $\Rightarrow G$ solvable. $\square$

Proposition 4.0.8. *Let G be a finite nonabelian simple group. Let p be a prime divisor of $|G|$.*

1. *If p is the smallest prime divisor, then G is divisible by p^3 or 12.*
2. *If $|G| = pm$ with $\gcd(p, m) = 1$, then for a Sylow p -subgroup P :*
 - (i) $C_G(P) \neq N_G(P) \neq G$;
 - (ii) $|N_G(P)/C_G(P)| \mid (p-1)$.

Proof. (1) Suppose $|G|$ is not divisible by p^3 . Then $p^2 \mid |G|$, otherwise a Sylow p -subgroup $P \cong C_p$ so that $N_G(P) = C_G(P)$, and G is p -nilpotent, contradiction since G is nontrivial simple. $\square$

Then $|P| = p^2$ and so $P \cong C_{p^2}$ or $C_p \times C_p$. The former would imply G is p -nilpotent, not possible. Hence $P \cong C_p \times C_p$ and $N_G(P) \neq C_G(P)$. So

$$N_G(P)/C_G(P) \simeq \text{Aut}(P) = \text{Aut}(C_p \times C_p) \simeq GL(2, p).$$

Now $|GL(2, p)| = p(p^2 - 1)(p - 1) = p(p + 1)(p - 1)^2$.

Let r be a prime divisor of $|N_G(P)/C_G(P)|$. Then $r \mid p(p + 1)$. And $r > p$ (since $P \cong C_p \times C_p$ so $P \leq C_G(P)$), so $p = 2$ and $p + 1 = 3$.

Hence $12 \mid |G|$.

(2) easy.

Example 4.0.9. *If $|G| = 5 \cdot 7 \cdot 13$, then G is cyclic.*

Let G_p denote a Sylow p -subgroup.

- $G_{13} \triangleleft G$ since $N_G(G_{13})/C_G(G_{13}) \leq \text{Aut}(G_{13}) = C_{12}$.
- G is 5-nilpotent.
- G is 13-nilpotent. Hence $G = G_7 \times G_{91}$.

Chapter 5

Week 5

Let G' be the commutator subgroup of G . Let $G'' = G^{(2)} = (G')'$ and $G^{(n)} = (G^{(n-1)})'$. We have

$$G \triangleright G' \triangleright G'' \triangleright \dots \triangleright G^{(n)} \triangleright \dots$$

Example 5.0.1. *Commutator subgroup G' of G :*

1. If G is simple and non-abelian, then G/G' is abelian and $G' \neq 1$, so $G' = G$.
2. $G = S_n$:

$$G' = \begin{cases} A_n & \text{if } n \geq 5 \\ A_4 & \text{if } n = 4 \end{cases}$$

Note that A_n is generated by all 3-cycles, since

$$(123) = (12)(13)(12)^{-1}(13)^{-1}.$$

If G is finite, then in the sequence

$$G \triangleright G' \triangleright G^{(2)} \triangleright \dots \triangleright G^{(n)} \triangleright G^{(n+1)} \triangleright \dots$$

either $G^{(n)} = G^{(n+1)} = G^{(n+2)} = \dots = G^{(\infty)}$ or $G^{(n)} = 1$.

Definition 5.0.2. 1. G is **solvable** if $G^{(n)} = 1$ for some n .

2. G is called a **perfect group** if $G = G'$.

Example 5.0.3. *Perfect groups:*

1. All non-abelian simple groups.
2. A_n with $n \geq 5$. For example, $(123)(345)(123)^{-1}(345)^{-1} = (143)$.
3. $\text{SL}_d(q)$ when nonsolvable, i.e., $d \geq 2$ and $(d, q) \neq (2, 2), (2, 3)$. Note that $Z(\text{SL}_d(q)) \cong \mathbb{Z}/(d, q-1)\mathbb{Z}$.

Proposition 5.0.4. *Properties of being solvable:*

1. If G is solvable, then so is each subgroup and each factor group of G .
2. If $N \triangleleft G$ such that N and G/N are solvable, then G is solvable.
3. p -groups are solvable (use (2) and induction on $|G|$, noting that $Z(G) \neq 1$ when G is nontrivial).

Lemma 5.0.5. *If P is a non-normal Sylow p -subgroup of G , then $P \subsetneq N_G(P)$.*

Definition 5.0.6. *Let $P < G$. A **double coset** of P in G is a set of the form*

$$PxP = \{ph_1h_2 \mid h_1, h_2 \in P\}.$$

Lemma 5.0.7. *The number of right cosets of P contained in PxP is equal to $|P|/|P^x \cap P|$.*

Proof. We have $PxP = \bigcup_{h \in P} P x h$. Note that $P x h_1 = P x h_2$ for $h_1, h_2 \in P$ if and only if $h_1 h_2^{-1} \in P^x \cap P$.

Decompose $G = \bigcup_{x \in G} PxP$ (note that for $x, y \in G$, either $PxP = PyP$ or $PxP \cap PyP = \emptyset$). We have:

- $P1P = P$ has a unique coset of P .
- $|G| = |P1P| + |Px_1P| + \cdots + |Px_tP|$.

Consider the number of cosets of P on both sides. The LHS is divisible by $|P|$. On the RHS, $|Px_iP| = |P|/|P^{x_i} \cap P|$ is either divisible by $|P|$ or equals 1. Then for some x_j , we have $|P^{x_j} \cap P| = |P|$, so $P^{x_j} \cap P = P$ and $P^{x_j} = P$, i.e., $x_j \in N_G(P)$. Thus $P \subsetneq N_G(P)$. $\square$

Theorem 5.0.8. *Let G be a p -group.*

1. If $H < G$, then $H < N_G(H)$.
2. If H is a maximal subgroup of G , then $H \triangleleft G$ and $|G|/|H| = p$ (consider G/H).
3. G is solvable (find H maximal in G , use induction).
4. $Z(G) > 1$.
5. If N is a normal subgroup of G and $|N| = p$, then $N \leq Z(G)$.

Proof of (5). By the N/C lemma, $\square$

$$G/C_G(N) = N_G(N)/C_G(N) \leq \text{Aut}(N) = \mathbb{Z}_{p-1}.$$

Then $G = C_G(N)$, so $N \leq Z(G)$.

Corollary 5.0.9. *A group of order p^2 with p prime is abelian.*

Proof. $Z(G)$ is nontrivial. □

Question: What about $|G| = p^3$? (Note: D_8, Q_8 are not abelian.)

Theorem 5.0.10 (Structure of G when $|G| = p^3$). *Let $|G| = p^3$. Then either G is abelian (3 types) or*

1. *for $p = 2$, $G \cong D_8$ or Q_8 ;*
2. *for $p \geq 3$, $G \cong \mathbb{Z}_{p^2} \rtimes \mathbb{Z}_p$ or $G = \langle a, b, c \mid a^p = b^p = c^p = 1, [a, b] = c \rangle$ (exponent p).*

Proof. Assume G is nonabelian.

(1) Suppose G has an element a of order p^2 .

By the above theorem, $\langle a \rangle \triangleleft G$. Then there is $b \in G \setminus \langle a \rangle$ and $b^p \in \langle a \rangle$ (since $b\langle a \rangle$ is of order p in $G/\langle a \rangle$).

If $b^p = 1$, then $G = \langle a \rangle \rtimes \langle b \rangle$ as in (1) for $p = 2$, (2) for $p \geq 3$.

Suppose $|b| = p^2$. Then $b^p = a^{lp}$, $a^b = a^\lambda$ for some integer λ . If $p = 2$, then $a^b = a^\lambda = a^{-1}$ and $b^a = b^\lambda = b^{-1}$. We have $G = \langle a, b \mid a^4 = b^4, a^b = a^{-1}, b^a = b^{-1} \rangle$, with unique element of order 2: $a^2 = b^2$.

Suppose p is odd. Then $a^b = a^{1+kp}$ for some k with $1 \leq k \leq p-1$ (from $a^{b^p} = a = a^{\lambda^p} \Rightarrow \lambda^p \equiv 1 \pmod{p^2} \Rightarrow \lambda^p \equiv 1 \pmod{p} \Rightarrow \lambda^p \equiv \lambda \pmod{p} \Rightarrow \lambda \equiv 1 + kp$).

Claim: $(ba^{-l})^p = 1$.

We compute:

$$\begin{aligned} ba^{-l} \cdot ba^{-l} &= b^2 b^{-l} a^{-l} b a^{-l} = b^2 a^{-2l-lkp}, \\ (ba^{-l})^p &= b^p a^{-pl-lkp^2} = 1. \end{aligned}$$

Let $b' = ba^{-l}$ as in (2).

(2) Suppose G does not have elements of order p^2 .

Then each non-identity element of G is of order p . Note: p must be odd, since if every element of G is of order 2, then G is abelian.

Consider G' . Since $Z(G)$ is nontrivial, pick $z \in Z(G)$, then $\mathbb{Z}_p = \langle z \rangle \triangleleft G$. We have $|G/\langle z \rangle| = p^2 \Rightarrow G/\langle z \rangle$ is abelian by the above discussion.

Hence $G' \leq \langle z \rangle \Rightarrow G' = \mathbb{Z}_p = \langle [a, b] \rangle$.

Let $c = [a, b]$. Then $a^p = b^p = c^p = 1$, $c = [a, b]$ as in (2). (Note: $\langle a, b \rangle \leq G$, $|\langle a, b \rangle| \neq p$ and $\neq p^2$ since a, b do not commute. So $\langle a, b \rangle = G$.) □

Automorphism groups of groups of order p^3 :

- $\text{Aut}(D_8) \simeq D_8$
- $\text{Aut}(Q_8) \simeq S_4$

- $\text{Aut}(\mathbb{Z}_{p^2} \rtimes \mathbb{Z}_p) \simeq P \rtimes \mathbb{Z}_{p-1}$
- $\text{Aut}(p_+^{1+2}) \simeq GL(2, p)$

Example 5.0.11. Let $G = P_-^{1+2}$ or P_+^{1+2} . Then $G' = Z(G) = \mathbb{Z}_p$.

Let $\tau \in \text{Aut}(G)$. Then $(G')^\tau = G'$ ($(z(G))^\tau = z(G)$), and τ induces an automorphism $\bar{\tau}$ of $G/G' \simeq \mathbb{Z}_p \times \mathbb{Z}_p$.

1. $G = \langle a \rangle \rtimes \langle b \rangle = \mathbb{Z}_{p^2} \rtimes \mathbb{Z}_p$, then $G' = \langle a^p \rangle$.

This is because $a^b = a^i \Rightarrow a^{b^p} = a^{ip} = a \Rightarrow i = 1 + lp \Rightarrow (a^p)^b = a^p$.

Then $G/G' = \langle \bar{a}, \bar{b} \rangle = \mathbb{Z}_p \times \mathbb{Z}_p$ (abelian) such that $\bar{a}^\tau \in \langle \bar{a} \rangle$. If $(|z|, p) = 1$, then $|\bar{z}| \mid p - 1$.

2. $G = P_+^{1+2}$, then $\bar{z} \in \text{Aut}(\langle \bar{a} \rangle \times \langle \bar{b} \rangle)$.

τ fixes $G' = \langle [x, y] \mid x, y \in G \rangle$ bilinear form.

$\tau \in Sp(2, p) \simeq SL(2, p)$, extension by scalar multiplication.

Chapter 6

Week 6

Free Groups.

Definition 6.0.1 (Free Group). Let $\{x_1, \dots, x_n\}$ be a set and $\{x'_1, \dots, x'_n\}$ such that $\{x_1, \dots, x_n\} \cap \{x'_1, \dots, x'_n\} = \emptyset$. Let $S = \{x_1, \dots, x_n, x'_1, \dots, x'_n\}$.

Let X be the set of all words (finite sequences) composed of the elements of S . For $u, v \in X$,

$$u = x_{i_1} \cdots x_{i_r}, \quad v = y_{j_1} \cdots y_{j_t},$$

define:

$$1. \quad u \cdot v = x_{i_1} \cdots x_{i_r} y_{j_1} \cdots y_{j_t}.$$

Another way of definition via “reduced words”: x_i and x'_i are not adjacent. If $u = u_1 x_i$ and $v = x'_i v_1$, then $uv = u_1 x_i x'_i v_1$. This identification is an equivalence relation, which gives rise to an equivalence partition of X . Let $[uv]$ denote the class containing uv . Then $[u]$ denotes the class containing u , and $[u][v] = [uv]$.

Then $\bar{X} = \{[u] \mid u \in X\}$ forms a group, called a free group generated by $x_1, \dots, x_n$, denoted by $F_n = \langle x_1, \dots, x_n \rangle$. The integer n is called the rank of F_n .

Example 6.0.2. $X = \langle x_1, x_2 \rangle$. If $g \in X$, then

$$g = x_1^{m_1} x_2^{m_2} x_1^{m_3} \cdots x_2^{m_t} \cdots \quad \text{or} \quad x_2^{m_1} x_1^{m_2} \cdots x_1^{m_t} \cdots$$

where $|m_i| \geq 1$.

Example 6.0.3. 1. $X = \langle x_1, x_2 \mid x_1^2 = 1, x_2^2 = 1 \rangle$ infinite dihedral group.

2. $X = \langle x_1, x_2 \mid x_1^2 = x_2^2 = 1, |x_1 x_2| = m \rangle$ D_{2m} .

3. $X = \langle x, y \mid |x| = m, |y| = n, |xy| = l \rangle$ triangle group.

Triangle Groups $T_{m,n,l}$

Consider $(\frac{1}{m} + \frac{1}{n} + \frac{1}{l})$:

- $> \pi$: spherical
- $= \pi$: Euclidean
- $< \pi$: hyperbolic

WLOG, assume $m \leq n \leq l$. Then if $G \neq \mathbb{Z}_2$, $l > 1$.

Case 1: $\frac{1}{m} + \frac{1}{n} + \frac{1}{l} > 1$.

For $m = 2$, $n = 3 \Rightarrow l = 3$ or 4 or 5 .

Further analysis shows $G = A_4$, S_4 , or A_5 .

Case 2:

$$\frac{1}{m} + \frac{1}{n} + \frac{1}{l} = 1$$

The solutions are $(m, n, l) = (3, 3, 3)$, $(2, 3, 6)$, or $(2, 4, 4)$.

The group $G = \langle x, y \mid x^m = y^n = (xy)^l = 1 \rangle$ is infinite and solvable.

These correspond to the three regular tilings of the plane:

- $(3, 3, 3)$: equilateral triangles
- $(2, 3, 6)$: regular hexagons (dual of equilateral triangles)
- $(2, 4, 4)$: squares

(These are the only three regular polygons that can tile the plane.)

Case 3:

$$\frac{1}{m} + \frac{1}{n} + \frac{1}{l} < 1$$

The group $G = \langle x, y \mid x^m = y^n = (xy)^l = 1 \rangle$ is infinite and nonsolvable.

Example 6.0.4. Consider $S_n = \langle (12), (12 \cdots n) \rangle$. Then

$$S_n = \langle x, y \mid x^2 = y^n = 1, (xy)^{n-1} = 1, \dots \rangle.$$

There must be additional relations. Otherwise, by the above discussion,

$$1 > \frac{1}{2} + \frac{1}{n} + \frac{1}{n-1},$$

which would imply S_n is infinite, a contradiction.

Theorem 6.0.5. Every finitely generated group is a homomorphic image of a free group. Let $G = \langle g_1, \dots, g_n \rangle$. Then F_n is homomorphic to G .

Proof. Define $\varphi : F_n \rightarrow G$ by $x_i \mapsto g_i$ and $x_i^{-1} \mapsto g_i^{-1}$. Let $N = \ker \varphi = \{x \mid \varphi(x) = 1\}$. Then $F_n/N \simeq G$. $\square$

We call $\{g_1, \dots, g_n\}$ the *generating elements* (or *generators*) and $\{\varphi(x) = 1 \mid x \in N\}$ the *generating relations*.

For example,

$$D_{2m} = \langle x, y \mid x^2 = y^2 = (xy)^m = 1 \rangle = \langle a, y \mid a^m = y^2 = 1, a^y = a^{-1} \rangle.$$

Minimal Normal Subgroups.

Let $\Omega = \{1, 2, \dots, n\}$. If $H < \text{Sym}(\Omega) \simeq S_n$ is intransitive, then $H \leq S_K \times S_{n-K}$ for some proper subset $K \subset \Omega$.

Let $G \leq \text{Sym}(\Omega) = S_n$.

Definition 6.0.6. A group G is called a *quasi-primitive group* if all its nontrivial normal subgroups are transitive.

Example 6.0.7. Consider $G = D_{2n} < S_n$ with generators $x = (1\ 2\ \dots\ n)$ and $y = (1\ n)(2\ n-1)\dots$. When n is even, $\{b, a^2b, \dots, a^{n-2}b\} \triangleleft G$, but it has $\frac{n}{2}$ elements, so it cannot be transitive.

Proposition 6.0.8. Let G be a quasi-primitive group. Let M be a minimal normal subgroup of G . Then:

1. M is transitive. (For an arbitrary group G , this is not always true; e.g., $G = \{(12)(34), (13)(24), (14)(23), 1\} < A_4$ has minimal normal subgroup $\langle (12)(34) \rangle$ which is not transitive.)
2. $M = T^l \simeq T_1 \times \dots \times T_l$ where $l \geq 1$ and $T_1 \simeq \dots \simeq T_l \simeq T$ are simple.

Lemma 6.0.9. If $M \triangleleft G$ is transitive, then $G = MG_w$ for any $w \in \Omega$.

Proof. Given $g \in G$, let $w' = w^g$. Since M is transitive on Ω , there exists $m \in M$ such that $w^m = w^g = w'$. Thus $m^{-1}g = y \in G_w$, i.e., $g = my \in MG_w$. $\square$

Lemma 6.0.10. Let G be a quasi-primitive group. Then G has at most two minimal normal subgroups. If M, N are two minimal normal subgroups, then $M \simeq N$, and both are nonabelian and regular.

Proof. That G has at most two minimal normal subgroups will be proved later.

If M, N are two minimal normal subgroups, then $M \cap N = \{1\}$. So $M \times N = MN \triangleleft G$ and M and N commute.

(1) **Regular.** Since N is transitive, for any $w, w' \in \Omega$, there exists $x \in N$ such that $w^x = w'$. Then

$$M_{w'} = M_{w^x} = x^{-1}M_w x = M_w$$

since M and N commute. So M_w fixes all points $w' \in \Omega$. Then $M_w = 1$ since M is faithful. Similarly $N_w = 1$.

(2) Nonabelian. If M or N is abelian, WLOG assume M is abelian. Since M and N commute, $\langle M, x \rangle = H$ for any $x \in N$ is strictly bigger than M and abelian. It is transitive, hence regular (since $H_w = H_{w'x} = x^{-1}H_{w'}x = H_{w'}$). But for regular groups A , for all $g, h \in \Omega$, there exists a unique $x \in A$ such that $g^x = h$. Then $|A| = |\Omega|$. Since M and N are regular, $|M| = |N| = |\Omega|$. Hence $|H| > |\Omega|$, a contradiction. Thus M and N are both nonabelian.

(3) $M \simeq N$. Let $X = M \times N \triangleleft G$. Then $X = MX_w = M \rtimes X_w$ and

$$X_w = \{(x, x') \mid x \in M, x' \in N\}$$

where x' is uniquely determined by x .

Define $\varphi : M \rightarrow N$ by $x \mapsto y$ if $(x, y) \in X_w$.

Prove that φ is a homomorphism. □

Chapter 7

Week 7

Lemma 7.0.1. *Let $G \leq \text{Sym}(\Omega)$ be transitive. Then $C_{\text{Sym}(\Omega)}(G)$ is semiregular.*

Proof. Let $C \in C_{\text{Sym}(\Omega)}(G)$. For $w, w' \in \Omega$, $\exists g \in G$ such that $w^g = w'$. Then $C_{w'} = C_{w^g} = C_w^g = g^{-1}C_w g = C_w$. So $C_w = 1$ and C is semiregular. $\square$

Corollary 7.0.2. *A transitive abelian permutation group G is regular.*

Proof. $G \leq C_{\text{Sym}(\Omega)}(G)$. $\square$

Theorem 7.0.3 (from last time). *Let G be a quasi-primitive group. Then G has at most two minimal normal subgroups. If G has two minimal normal subgroups M, N , then $M \simeq N$, and M, N are regular and nonabelian.*

Proof. M, N regular (use lemma). $X = MN = M \times N \Rightarrow M \leq C_{\text{Sym}(\Omega)}(N) \Rightarrow M$ is regular, so is N . Also $M \simeq N$.

For $w \in \Omega$, $X_w = \{(x, x') \mid x \in M, x' \in N\} \leq M \times N$.

Let $\varphi : M \rightarrow N$ by $x \mapsto x'$. This is surjective and injective by comparison. $\square$

Case: assume that G has a unique minimal normal subgroup N .

Lemma 7.0.4. *Let G have a unique minimal normal nonabelian subgroup N . Then $C_G(N) = \{1\}$, so $G \leq \text{Aut}(N)$ by N/C lemma.*

Proof. Since N is nonabelian, $N \neq C_G(N)$. Since $C_G(N) \triangleleft G$, it would contain a minimal normal subgroup if $C_G(N) \neq \{1\}$, contradiction. Thus $C_G(N) = \{1\}$. $\square$

Thus:

1. If $N = \mathbb{Z}^n$, then N is regular.
2. If N is nonabelian simple, then $N \triangleleft G \leq \text{Aut}(N)$.
3. Let $N = T^k$ where $k \geq 2$ and T is nonabelian simple:

- (i) N is regular $\Leftrightarrow N_w = 1$ TW (twisted wreath product).
- (ii) $N_w \cong T$ SD_k (simple diagonal).
- (iii) $N_w \simeq T^m$, $2 \leq m \leq \frac{k}{2}$ CD (compound diagonal).
- (iv) $1 \neq N_w \neq T^m$ for $m \geq 1$ PA (product action).

Theorem 7.0.5 (O’Nan–Scott–Praeger). *Let G be a quasi-primitive group. Then one of the following holds:*

1. G has exactly two minimal normal subgroups $M \simeq N$ HS (holomorph simple).
2. G has two minimal normal subgroups $M \simeq N \simeq T^k$ with $k \geq 2$ HC type (holomorph compound).
3. G has a unique minimal normal abelian subgroup HA type (holomorph affine).
4. G has a unique minimal normal subgroup N , simple and nonabelian AS type (almost simple).
5. G has a minimal normal nonabelian regular subgroup.
6. Case (3)(ii) above.
7. Case (3)(iii).
8. Case (3)(iv).

Case 1: $G \leq \text{Sym}(\Omega)$ quasi-primitive. Let $M, N \triangleleft G$. Then $M \simeq N$, nonabelian, regular. So $M \times N$ and $|\Omega| = |M| = |N|$.

Fix $w \in \Omega$. For any $w' \in \Omega$, $\exists! x \in M$ such that $w^x = w'$.

$\Rightarrow$ There is a 1–1 correspondence between Ω and M by

$$\begin{aligned} w &\mapsto 1 \\ w' &\mapsto x'. \end{aligned}$$

Relabel:

$$\begin{aligned} \Omega &= \{w_1, w_2, \dots, w_n\} \quad \text{where} \quad \Omega \rightarrow M, w_i \mapsto h, \\ M &= \{1 = x_1, x_2, \dots, x_n\}, \quad w_i \mapsto x_i, w_i = w_1^{x_i}. \end{aligned}$$

Lemma 7.0.6 (Relationship between $\hat{g}$, $\tilde{g}$, right/left multiplication). *Let G be a*

finite group. Fix $g \in G$. Define permutations on G :

$$\begin{aligned}\hat{g} &: x \mapsto xg, \\ \tilde{g} &: x \mapsto g^{-1}x, \\ \bar{g} &: x \mapsto gx^{-1}.\end{aligned}$$

Then:

$$\begin{aligned}\hat{G} &= \{\hat{g} \mid g \in G\}, \\ \tilde{G} &= \{\tilde{g} \mid g \in G\}, \\ \bar{G} &= \{\bar{g} \mid g \in G\}.\end{aligned}$$

1. $\hat{G}, \tilde{G} \leq \text{Sym}(G)$, and both are regular.
2. $\langle \hat{G}, \tilde{G} \rangle = \hat{G}\tilde{G} = \hat{G} \cdot \tilde{G}$, central product, with $\hat{G} \cap \tilde{G} = Z(\hat{G}) = Z(\tilde{G})$.
3. Let $X = \hat{G}\tilde{G}$, $X_1 = \tilde{G}$, then $X = \hat{G} : \tilde{G} = \tilde{G} : \hat{G}$.

Definition 7.0.7 (Central Product $M \circ N$ or $M * N$). Let $C_1 \leq Z(M)$, $C_2 \leq Z(N)$ with $C_1 \simeq C_2$ via isomorphism $\varphi: C_1 \rightarrow C_2$. Set

$$C = \{(c, c^\varphi) \mid c \in C_1\} \simeq C_1.$$

The central product is defined as

$$M \circ N = M \times N / C.$$

Usually when C is maximal, i.e., $Z(M) \simeq Z(N) \simeq C$, this is called the standard central product.

Example 7.0.8. 1. $Q_8 \times Q_8$: Here $Z = \{1, -1\} \cong Z_2$. Let $Z = \langle c_1 \rangle = \langle c_2 \rangle$. Then

$$Q_8 \times Q_8 / \langle (c_1, c_2) \rangle = Q_8 \circ Q_8,$$

with $|Q_8 \circ Q_8| = 32$.

2. $Q_8 \circ Z_4 = Q_8 \times Z_4 / \langle (c_1, c_2) \rangle$.

3. Let $M = Z_8 : Z_2$ with $\langle a \rangle : \langle z \rangle$ and $a^z = a^5$. Then $(a^2)^z = a^{10} = a^2$, so $a^2 \in Z(M) = \langle a^2 \rangle \cong Z_4$.

Let $N = \langle b \rangle : \langle y \rangle$ with $b^y = b^5$.

Central products. Let $C_2 \in Z(N) = Z_4$. If $|c_1| = |c_2| = 4$, then

$$M \times N / \langle (c_1, c_2) \rangle = M \cdot N$$

is a central product. If $|k_1| = |k_2| = 2$, then

$$M \times N / \langle (c_1, c_2) \rangle = M \circ N.$$

Lemma 7.0.9. 1. $\hat{G}$ is a group: if $\hat{g}, \hat{h} \in \hat{G}$, then $\hat{g}\hat{h} = \widehat{gh} \in \hat{G}$. Similarly $\tilde{G}$ is a group, since

$$x^{\hat{g}\tilde{h}} = (x^{\hat{g}})^{\tilde{h}} = (g^{-1}x)^{\tilde{h}} = h^{-1}g^{-1}x = x^{\tilde{g}h}.$$

Moreover, for any $x, y \in \Omega(= G)$, $\hat{g} = x^{-1}y$ maps x to y , so $\hat{G}$ is transitive. Thus G is regular since $|\hat{G}| = |G|$.

2. Let $\hat{g} \in \hat{G}$ and $\tilde{h} \in \tilde{G}$.

Claim: $\hat{g}\tilde{h} = \tilde{h}\hat{g}$ in $\text{Sym}(\Omega)$.

For $x \in G$, we have

$$\begin{aligned} x^{\hat{g}\tilde{h}} &= (x^{\hat{g}})^{\tilde{h}} = (xg)^{\tilde{h}} = h^{-1}xg, \\ x^{\tilde{h}\hat{g}} &= (x^{\tilde{h}})^{\hat{g}} = (h^{-1}x)^{\hat{g}} = h^{-1}xg. \end{aligned}$$

Hence $\hat{G}$ and $\tilde{G}$ commute and $\langle \hat{G}, \tilde{G} \rangle = \hat{G}\tilde{G}$. Moreover, $\hat{G} \cap \tilde{G} = 1$ since $\hat{G}$ and $\tilde{G}$ commute.

The map $\varphi: M \times N \rightarrow MN$ given by $(m, n) \mapsto mn$ is a homomorphism since $\hat{G}$ and $\tilde{G}$ commute. We have

$$\ker \varphi = \{(\hat{g}, \tilde{g}) \mid g \in Z(G)\}, \quad \hat{g} \in Z(M).$$

3. Let $X = \langle \hat{G}, \tilde{G} \rangle = \hat{G} \circ \tilde{G}$. Then

$$X_w = \{(\hat{g}, \tilde{g}) \mid g \in G\}$$

and $x^{(\hat{g}, \tilde{g})} = g^{-1}xg$, where $w = 1 \in G$.

Since $\hat{G}$ is regular, $\hat{G} \cap X_w = 1$ and $\hat{G} \triangleleft X$. Thus $X_1 = \tilde{G}$ and

$$X = \hat{G} \rtimes X_w = \hat{G} \rtimes \tilde{G}.$$

So $\hat{G}\tilde{G} = \hat{G} \rtimes \tilde{G} = \tilde{G} \rtimes \hat{G}$ (internal semidirect product).

In particular, if $Z(G) = 1$, then $\hat{G} \times \tilde{G} \cong G \times G$.

Return to case (1). Identify Ω with M . Then $\hat{M}$ acts regularly on M by right multiplication. Let $X = M \times N < \text{Sym}(\Omega)$. Then

$$X_w = \{(g, g') \mid g \in M\}.$$

Define $\varphi: M \rightarrow N$ by $g^\varphi = g'$. Then $X = \hat{M} \rtimes M$ acts on M . We aim to prove the two actions are equivalent.

Recall $\tilde{\omega} \mapsto 1\mu$ and $X_w = \{(g, g') \mid g \in M\}$.

Let $X_1 = \{(\hat{g}, \check{g}) \mid g \in M\}$. Let $\varphi : M \rightarrow N$ be such that $g' = g^\varphi$.

We have $(g, 1) \mapsto (\hat{g}, 1)$ and $M \times \{1\} \rightarrow \{1\} \times M$.

Now $X = MN = M \times N = M \times M = \hat{M} \times \hat{M}$, so $M \equiv \hat{M}$ and $N \equiv \check{M}$.

Suppose $Z(a^2)^3 = 1$. Let $M < \text{Sym}(\Omega)$ be regular. Then $C_{\text{Sym}(\Omega)}(M) \geq N$, and by the lemma, $C_{\text{Sym}(\Omega)}(M) = N$.

Note that M is transitive and $C_{\text{Sym}(\Omega)}(M)$ is semiregular. Similarly, $C_{\text{Sym}(M)}(\hat{M}) = \check{M}$.

Lemma 7.0.10. *Let $M < \text{Sym}(\Omega)$ be regular. Then, identifying Ω with M , the action of M on Ω is equivalent to the action of $\hat{M}$ on M .*

Proof. Let $\Omega = \{w_1, w_2, \dots, w_n\}$ where $w_i = w_1^{g_i}$, and let $M = \{1 = g_1, g_2, \dots, g_n\}$.

Under the identification $w_i \mapsto g_i$, we have $w_i^g = w_1^{g_i g} \mapsto g_i g$. Thus $(g_i)^g = g_i g$, and the actions are equivalent. $\square$

Moreover, $N = C_{\text{Sym}(\Omega)}(M) = C_{\text{Sym}(M)}(\hat{M}) = \check{M}$.

Chapter 8

Week 8

Corollary 8.0.1. *Each regular action of a group X is equivalent to $\hat{X}$ acting on X .*

Proof. We only prove for X . Let $\varphi: X \rightarrow \Omega$. Define

$$\varphi: x_i \mapsto wx_i^{-1}, \quad x_v \mapsto x, \quad x_j \mapsto x_j.$$

This yields $x_i^{x_j} = x_j^{-1}x_i \mapsto x_i^{-1}x_j = (w^{x_i^{-1}})x_j$. Then φ is H -equivariant and ρ is an isomorphism. Hence the two actions are equivalent.

Definition 8.0.2 (Holomorph). *Let G be a group. The holomorph of G is $\text{Hol}(G) = G \rtimes \text{Aut}(G)$. Explicitly,*

$$\text{Hol}(G) = \{(g, \theta) \mid g \in G, \theta \in \text{Aut}(G)\}$$

with multiplication

$$(g_1, \theta_1)(g_2, \theta_2) = (g_1\theta_1(g_2), \theta_1\theta_2).$$

Case (1): Holomorph simple. Let $M \times N \triangleleft G \leq \text{Sym}(\Omega)$ where $M \simeq N \simeq T$ is simple. Then

$$M \rtimes \text{Inn}(M) \leq G \leq M \rtimes \text{Aut}(M).$$

Proof. For the left-hand side: $\hat{M} \times \hat{M} = M \times N \triangleleft G$, and $\hat{M} \cong M$ gives $M \rtimes \text{Inn}(M)$. For the right-hand side: $G = M \rtimes G_w \leq \text{Sym}(\Omega)$. Identifying M with Ω , we have

$$G = \hat{M} \rtimes G_1 \leq \text{Sym}(M).$$

Claim: $G_1 \leq \text{Aut}(M)$. For all $\theta \in G_1$, we have $\theta^{-1}\hat{m}\theta = \hat{m}'$ for some $m' \in M$. Then $\theta^{-1}\hat{m}\theta(1) = \hat{m}'(1)$ implies $\theta(m) = m'$. Let $\theta^{-1}\hat{m}\theta = \hat{m}'$ and $\theta^{-1}\hat{n}\theta = \hat{n}'$. Then $\theta^{-1}\hat{m}\hat{n}\theta = \hat{m}'\hat{n}'$, which yields $\theta(mn) = m'n' = \theta(m)\theta(n)$. Hence $G_1 \leq \text{Aut}(M)$.

Lemma 8.0.3 (Frattini Argument). *Let $H < G \leq \text{Sym}(\Omega)$ such that G is transitive on Ω . Then H is transitive on Ω if and only if $G = HG_w$ (i.e., for all $g \in G$, there exist $x \in H$ and $y \in G_w$ such that $g = xy$).*

Lemma 8.0.4. *Let $M \triangleleft G \leq \text{Sym}(\Omega)$ such that M is regular on Ω . Then $G = M \rtimes G_w$ where $w \in \Omega$.*

Case (2): Holomorph compound. Let $M \times N \triangleleft G \leq \text{Sym}(\Omega)$ where $M \simeq N = T^\ell$ with $\ell \geq 2$ and T nonabelian simple. Then

$$M \rtimes \widetilde{M} \leq G \leq M \rtimes \text{Aut}(M)$$

where $\text{Aut}(M) = \text{Aut}(T) \wr S_\ell$ (wreath product). Here $M \simeq N = T^\ell = T_1 \times \cdots \times T_\ell$ where $T \simeq T_1 \simeq \cdots \simeq T_\ell$, and G induces a transitive action on $\{T_1, \dots, T_\ell\}$.

Lemma 8.0.5. *A quasiprimitive permutation group of type HS, HC, or HA is primitive.*

Blocks and Block Systems

Let G be a transitive permutation group on Ω .

Definition.

1. A subset $B \subseteq \Omega$ is called a *block* if for any element $g \in G$, $B^g = \{\omega^g \mid \omega \in B\}$ satisfies either $B^g = B$ or $B^g \cap B = \emptyset$.
2. For a block B , the set $\mathcal{B} = \{B^g \mid g \in G\}$ is called a *block system* for G on Ω .

Properties of Blocks

Write $\mathcal{B} = \{B_1, B_2, \dots, B_m\}$.

1. Each B_i is a block.
2. $B_i \cap B_j = \emptyset$ for all $i \neq j$.
3. $|B_1| = \cdots = |B_m|$.
4. G induces a transitive action on $\mathcal{B}$.

Definition 8.0.6 (Primitive and Imprimitive). *If there exists a block B for G on Ω with $1 < |B| < |\Omega|$ (i.e., a nontrivial block), then G is said to be imprimitive on Ω . Otherwise, G is said to be primitive.*

Lemma 8.0.7. *A transitive group $G \leq \text{Sym}(\Omega)$ is primitive if and only if G_w is a maximal subgroup of G .*

Proof. ($\Rightarrow$) Assume otherwise: G is imprimitive on Ω , and let B be a nontrivial block. Let $\omega \in B$. Then

$$G_\omega \neq G_B = \{g \in G \mid B^g = B\}.$$

(There exists $\omega' \neq \omega$ in B and $g \in G$ such that $\omega^g = \omega'$; then $g \in G_B$ but $g \notin G_\omega$.) That is, $G_\omega < G_B < G$.

($\Leftarrow$) If $G_\omega < H < G$, then $B := \omega^H$ is a block.

1. $B^g \cap B = \emptyset$ or $B^g = B$: Let $g \in G$ and suppose $\delta \in B^g \cap B$. Then $\omega^{h_1 g} = \omega^{h_2}$ for some $h_1, h_2 \in H$, so $h_1 g h_2^{-1} \in G_\omega < H$. Since $h_1, h_2 \in H$, we have $g \in H$, hence $B^g = B$.
2. B is a nontrivial block: If H is transitive, then $G = HG_\omega = H$, a contradiction. Hence H is not transitive and $1 < |B| < |\Omega|$.

□

Proof of Lemma (Case HA). Let G be of type HA. Then $G = M \rtimes G_\omega$ where M is an abelian unique minimal normal subgroup.

Suppose G is imprimitive on $\Omega = M$. Then $G_\omega < H < G$, and $1 \neq H \cap M = M_0 < M$. (If $H \cap M = 1$, then $G = M \rtimes H$, contradiction.) Then $M_0 \triangleleft M$, $M_0 \triangleleft H$, and $M_0 \triangleleft \langle H, M \rangle = G$, contradicting M being minimal normal and M abelian. □

Example 8.0.8. A group can be quasiprimitive but not primitive. (Note that primitive implies quasiprimitive.)

Proof that primitive implies quasiprimitive: Let $H \triangleleft G$.

1. If $H < G_\omega$ for some $\omega \in \Omega$, then $H < G_{\omega^x}$ for any $x \in G$. Hence H fixes every element in Ω . Since G is faithful, $H = 1$.
2. If $H \not< G_\omega$ for any $\omega \in \Omega$, then HG_ω is a group strictly larger than G_ω , so $G = HG_\omega$.

Let G be a non-abelian simple group. Then $\hat{G}$ is a quasiprimitive group on G , but $\hat{G}$ is not primitive since $\hat{G}_\omega = 1$ is not maximal in $\hat{G}$.

Chapter 9

Week 9

HS type (HC type * (homework)).

Let $G = T:H$ with $\text{Inn}(T) \leq H \leq \text{Aut}(T)$.

- H can be G_w with $w = 1 \in T$ (or any other w').
- Claim: G is primitive on $\Omega(\equiv T)$ ($\Leftrightarrow G_w$ is maximal).

Proof. Suppose $H \leq K \leq G$. Then an element $g \in K \setminus H$ has the form $g = (t, h) \in G = T:H$ such that $t \neq 1$. Thus $(t, 1) = (t, h)(1, h^{-1}) \in K$.

For any element $x \in H$, we have $(1, x^{-1}) \in K$ and

$$(1, x^{-1})(t, 1) = (t^{x^{-1}}, x^{-1}) \in K,$$

so $(t^{x^{-1}}, x^{-1})(1, x) = (t^{x^{-1}}, 1) \in K$.

Let $M = \langle (t^x, 1) \mid x \in H \rangle \leq T \times \{1\}$.

Since $H \geq \text{Inn}(T)$, the action of T on M by conjugation fixes M , so $M \trianglelefteq T$. Since T is simple, $\langle M, H \rangle = \langle T, H \rangle = T:H = G \leq K$. So H is a maximal subgroup of G and G is primitive on Ω . $\square$

Corollary 9.0.1. *This provides a way to construct primitive groups. For T simple, let $G = T \times T = T:\tilde{T} = T:\text{Inn}(T)$ (since T simple, $Z(T)$ is trivial, hence semidirect product).*

Notation

Let $G \leq \text{Sym}(\Omega)$, $|\Omega| = n$.

- For $\Delta \subseteq \Omega$ and $g \in G$, let $\Delta^g = \{\delta^g \mid \delta \in \Delta\}$.
- For $w_1, w_2 \in \Omega$, write $w_1 \sim w_2$ if $w_1^g = w_2$ for some $g \in G$.
- $w^G = \{w^g \mid g \in G\}$ is the *orbit* of w .
- $G_w = \{g \in G \mid w^g = w\}$ is the *stabilizer* of w .

Subgroups and Actions on Subsets

For $\Delta \subseteq \Omega$, let

$$G_\Delta = \{g \in G \mid \Delta^g = \Delta\}$$

and

$$G_{(\Delta)} = \{g \in G \mid w^g = w \text{ for all } w \in \Delta\}.$$

Then $G_{(\Delta)} \trianglelefteq G_\Delta$ and G_Δ acts on Δ with kernel $G_{(\Delta)}$.

We have:

- $g^{-1}G_wg = G_{w^g}$
- $g^{-1}G_\Delta g = G_{\Delta^g}$
- $g^{-1}G_{(\Delta)}g = G_{(\Delta^g)}$

The group G_Δ acting on Δ induces a permutation group on Δ , denoted by G_Δ^Δ , and $G_\Delta^\Delta \simeq G_\Delta/G_{(\Delta)}$.

Fixed Points

- $\text{Fix}_\Omega(G) = \{w \in \Omega \mid w^g = w \text{ for all } g \in G\}$, the set of fixed points of G on Ω .
- $\text{Fix}_\Omega(g) = \{w \in \Omega \mid w^g = w\}$, the set of fixed points of g on Ω .

Definition 9.0.2 (Semiregular). *G is said to be semiregular on Ω if $G_w = \{1\}$ for each point $w \in \Omega$.*

- $|G| = \text{length of each orbit} \Rightarrow |G| \mid |\Omega|$.
- If G is transitive and semiregular, then G is *regular* on Ω and $|G| = |\Omega|$.

Definition 9.0.3 (Coset Action). *Let G be a group and $H \leq G$. Let $\Omega = [G : H] = \{Hg \mid g \in G\}$. Define an action of G on Ω by*

$$(Hg) \cdot x = H(gx) \in [G : H] \quad \text{for all } Hg \in \Omega \text{ and } x \in G,$$

i.e., $x: Hg \mapsto Hgx$.

- A coset action is transitive: $(Hg_1)^{g_2^{-1}} = Hg_2$.
- The kernel of this action is $G(\Omega) = \bigcap_{g \in G} g^{-1}Hg$.

Denote $G(\Omega)$ by M . Then $M = \bigcap_{g \in G} g^{-1}Hg \triangleleft G$, $M \leq H$, and M is the largest normal subgroup of G contained in H , called the *core* of H in G , denoted by $\text{Core}(H)$.

Proposition 9.0.4. $G/\text{Core}_G(H) \simeq G^\Omega$ is a transitive permutation group on $\Omega = [G : H]$.

- If $\text{Core}_G(H) = 1$, then the action is said to be *faithful* (H is core-free in G). Otherwise the action is unfaithful.

Theorem 9.0.5. Each transitive permutation group is equivalent to a faithful coset action.

Proof. Let G be a transitive permutation group on Ω , i.e., $G \leq \text{Sym}(\Omega)$ is transitive. Let $H = G_w$ with $w \in \Omega$ and let $\Sigma = [G : H]$. Then for $C = Hg$ and $x \in G$, $C^x = (Hg)^x = Hgx \in \Sigma$.

$$\begin{aligned}\Omega : & w, w^{g_1}, w^{g_2}, \dots, w^{g_n} \\ \Sigma : & H, Hg_1, Hg_2, \dots, Hg_n\end{aligned}$$

For $x \in G$:

$$\begin{aligned}x : w^{g_i} &\mapsto w^{g_j} \\ Hg_i &\mapsto Hg_j.\end{aligned}$$

H is core-free in G , i.e., $\text{Core}_G(H) = 1$, meaning H does not contain a non-identity normal subgroup of G .

Suppose $N \leq H$ and $N \triangleleft G$. Then $N = N \cdot w$ since $N \leq H$. For $x \in G$, $Nw = N = x^{-1}Nx = x^{-1}N_w x = N_w x$, so N fixes every point of Ω . We then must have $N = 1$ as $G \leq \text{Sym}(\Omega)$. Thus $\text{Core}_G(H) = 1$ and G acts on Σ faithfully. $\square$

Definition 9.0.6 (Equivalence of actions). Let G act on Ω and Σ transitively and faithfully. Then G^Ω and G^Σ are said to be equivalent if there is a bijection $\varphi : \Omega \rightarrow \Sigma$ such that for any $w \in \Omega$ and $g \in G$,

$$\varphi(w^g) = \varphi(w)^g.$$

Example 9.0.7. Are the following actions equivalent?

$$\begin{aligned}G &= S_4 \text{ on } \{1, 2, 3, 4\} \\ H_1 &= \langle (12) \rangle \\ H_2 &= \langle (12)(34) \rangle \\ H_3 &= \langle (13) \rangle\end{aligned}$$

Let $\Omega_i = [G : H_i]$. Compare G^{Ω_1} , G^{Ω_2} , and G^{Ω_3} .

(1) G^{Ω_1} and G^{Ω_2} : Note that $H_1(12) = H_1$. If $H_1 \mapsto H_2g$ for some $g \in G$, then we

must have $H_2g = H_2g(12)$, which means either

$$\begin{cases} g(12) = g, & \text{or} \\ g(12) = (12)(34)g. \end{cases}$$

Both are not possible. Hence G^{Ω_1} and G^{Ω_2} are not equivalent.

(2) G^{Ω_1} and G^{Ω_3} : We may take $H_1 \mapsto H_3(23)$. Define $\varphi: [G : H_1] \rightarrow [G : H_3]$ by

$$\varphi(H_1g) = H_3(23)g.$$

One checks that φ is well-defined and gives an equivalence.

Indeed, if $H_1g_1 = H_1g_2$, then $g_1g_2^{-1} \in H_1 = H_3$. Hence

$$(23)H_3(23)g_1 = (23)H_3(23)g_2 \Rightarrow H_3(23)g_1 = H_3(23)g_2.$$

Theorem 9.0.8 (Criterion for equivalence). *The transitive actions G^Ω and G^Σ are equivalent if and only if G_w and G_δ are conjugate in G .*

Proof. ($\Leftarrow$) The defined map is well-defined.

$$(\Rightarrow) \Omega \rightarrow \Sigma: g^{-1}G_\delta g \subset G_w.$$

$$\Sigma \rightarrow \Omega: g^{-1}G_w g \subset G_\delta.$$

□

Chapter 10

Week 10

Exercise 10.0.1. Characterize quasiprimitive permutation groups $G \leq \text{Sym}(\Omega)$ such that G_ω has r orbits on Ω where $r = 2, 3, 4, 5$. If G is of holomorph simple type, then $r \geq ?$

Theorem 10.0.2 (Burnside). A 2-transitive permutation group is affine (holomorph affine) or almost simple (AS). For all such groups, $\text{rank}(G) = 2$.

Definition 10.0.3 (Rank). Let $G \leq \text{Sym}(\Omega)$ be transitive. Then G induces an action on $\Omega \times \Omega = \{(\omega_1, \omega_2) \mid \omega_1, \omega_2 \in \Omega\}$ by

$$(\omega_1, \omega_2)^g = (\omega_1^g, \omega_2^g) \quad \forall g \in G.$$

Then G acts intransitively and partitions $\Omega \times \Omega$ into G -orbits.

The number of G -orbits is called the rank of G on Ω , denoted by $\text{rank}(G)$.

- $\{(\omega, \omega) \mid \omega \in \Omega\}$ is an orbit, so $\text{rank}(G) \geq 2$.
- If $\text{rank}(G) = 2$, then G is called 2-transitive.
- A 2-transitive group is primitive.
- If G is 2-transitive on Ω , then:
 - (i) G_ω is faithful and transitive on $\Omega \setminus \{\omega\}$.
 - (ii) $|\Omega| - 1$ divides $|G_\omega|$.
 - (iii) If $N \triangleleft G$, then N_ω is half-transitive on $\Omega \setminus \{\omega\}$, i.e., all N_ω -orbits have the same size.

Note: (iii) uses the following lemma with $N_\omega = N \cap G_\omega$ and G_ω where G_ω is transitive on $\Omega \setminus \{\omega\}$.

Lemma 10.0.4. Let $G \leq \text{Sym}(\Omega)$ be transitive. If $1 \neq N < G$ is intransitive, then the N -orbits on Ω form a nontrivial block system and N is half-transitive.

Proof. Let Δ_1, Δ_2 be two N -orbits. Choose $\delta_i \in \Delta_i$ with $i = 1, 2$. Then $\delta_1^g = \delta_2$ for some $g \in G$.

Write $\Delta_1 = \{\delta_1^{x_1}, \delta_1^{x_2}, \dots, \delta_1^{x_t}\}$ with $x_i \in N$. Then

$$(\delta_1^{x_i})^g = \delta_1^{gg^{-1}x_i g} = (\delta_1^g)^{g^{-1}x_i g} = \delta_2^{g^{-1}x_i g} \in \Delta_2.$$

So $\Delta_1^g \subseteq \Delta_2$. Similarly $\Delta_2^{g^{-1}} \subseteq \Delta_1$, so $|\Delta_1| = |\Delta_2|$. $\square$

Corollary 10.0.5. *If G is primitive, then G is quasiprimitive.*

Proof of Theorem. Let $G \leq \text{Sym}(\Omega)$ be 2-transitive. Let $N \triangleleft G$ such that $N = T^k$ with $k \geq 2$ and T nonabelian simple.

Lemma 10.0.6. *N does not have a proper normal subgroup which is regular on Ω (i.e., G is not HS, HC, SD, or CD).*

Proof. Suppose $M \triangleleft N$ is regular on Ω . Then $|\Omega| = |M|$ and $N_\omega \neq 1$ (since $M < N$) is half-transitive on $\Omega \setminus \{\omega\}$. Let m be the size of an orbit of N_ω on $\Omega \setminus \{\omega\}$. Then $m \mid |\Omega \setminus \{\omega\}|$ and $m \mid |N_\omega|$. Thus $m \mid \gcd(|M| - 1, |N_\omega|)$.

Now $|M| = |T|^l$ for $l < k$, and $N = M \rtimes N_\omega$ so that $|N_\omega| = |T|^{k-l}$. Thus

$$m \mid \gcd(|T|^l - 1, |T|^{k-l}) = 1 \Rightarrow m = 1,$$

a contradiction (since N_ω is faithful on $\Omega \setminus \{\omega\}$). $\square$

Lemma 10.0.7. *N is not regular on Ω (i.e., G is not TW).*

Proof. Suppose N is regular on Ω . Then $G = N \rtimes G_\omega$ and $N \cap G_\omega = N_\omega = 1$. Hence $G = N \rtimes G_\omega$, and Ω can be identified with N with $\omega \mapsto 1$, so that G_ω acts on N by conjugation:

$$\omega^n \mapsto n; \quad \omega^{ng} = \omega^{g^{-1}ng} \mapsto g^{-1}ng = n^g.$$

The action of G_ω partitions N into orbits and fixes 1.

Note that elements of N in the same orbit have the same order.

Clearly $|N|$ has at least 3 prime divisors r, s, t . (For $N = T^k$ where $k \geq 2$ and T nonabelian simple, N is not solvable, while groups of odd order are solvable, and groups of order $p^a q^b$ are solvable by Burnside.)

Then there exist $x, y, z \in N$ such that $|x| = r, |y| = s, |z| = t$. Thus $x^{G_\omega}, y^{G_\omega}, z^{G_\omega}$ are 3 distinct orbits of G_ω on N . Since $\text{rank}(G) = \#$ of G_ω -orbits on Ω , we have $\text{rank}(G) \geq 4$, a contradiction. $\square$

Remark 10.0.8. *An example with $\text{rank}(G) = 4$: If $N = T = A_5$, then $(r, s, t) = 2, 3, 5$. With $G_\omega = S_5$, we have $\text{rank}(G) = 4$ (HS type).*

Lemma 10.0.9.

Let G be 2-transitive and let $N \trianglelefteq G$ be a minimal normal subgroup such that $N = T^k$ with $k \geq 2$ and T nonabelian simple. Then N is the unique minimal normal subgroup. Now G acts on Ω by

$$(\delta_1, \dots, \delta_k)^{(z_1, \dots, z_k)\pi} = (\delta_1^{z_1\pi^{-1}}, \dots, \delta_k^{z_k\pi^{-1}})$$

(sometimes called type PA, i.e., G acts on Ω in product action).

Proof. $N_\omega \neq 1$. Write $N = T_1 \times \dots \times T_k$ where $T_1 \simeq \dots \simeq T_k$. Let H_i be the projection image of N_ω into T_i : for $g \in N_\omega < T_1 \times \dots \times T_k$, write $g = (t_1, \dots, t_k)$ where $t_i \in T_i$, and project to t_i .

Let $M_i = \prod_{j \neq i} T_j$. Then $M_i \triangleleft N$ and $N/M_i \cong T_i$. Then $H_i = N_\omega / (N_\omega \cap M_i) \cong N_\omega M_i / M_i$, and $H_i \leq T_i$ with $H_1 \simeq H_2 \simeq \dots \simeq H_k$. If not, any orbit is a proper smaller normal subgroup.

Since $G = NG_\omega$ is transitive on $\{T_1, \dots, T_k\}$ (as $T_i^g \triangleleft N \Rightarrow T_i^g = T_j$ for some j , for all $g \in G$), and N acts trivially on $\{T_1, \dots, T_k\}$ by conjugation, so G_ω is transitive on $\{T_1, T_2, \dots, T_k\}$. Also, G_ω normalizes N_ω , so G_ω is transitive on $\{H_1, \dots, H_k\}$ by conjugation.

Then $N_\omega \leq H_1 \times \dots \times H_k \leq N$ and $K = H_1 \times \dots \times H_k$ is normalized by G_ω . Thus $G_\omega \leq KG_\omega < G$. Since G is primitive, G_ω is maximal. Then $G_\omega = KG_\omega \Rightarrow K = N_\omega \leq G_\omega$ (so $K \leq N$ and $K \leq G_\omega$).

Consider $N = T_1 \times \dots \times T_k$ acting on Ω . Identify Ω with $[N : N_\omega]$:

$$\Omega = [N : N_\omega] = [T_1 \times \dots \times T_k : H_1 \times \dots \times H_k] = [T_1 : H_1] \times \dots \times [T_k : H_k].$$

with N acting on Ω defined by

$$w(t_1, \dots, t_k) = (H_1 x_1, \dots, H_k x_k)^{(t_1, \dots, t_k)} = (H_1 x_1 t_1, \dots, H_k x_k t_k)$$

where $w = (H_1 x_1, \dots, H_k x_k)$.

We have $G_\omega \leq (L_1 \times \dots \times L_k) : S_k$ where $H_i \leq L_i \leq \text{Aut}(T_i)$, and

$$G \leq (\text{Aut}(T_1) \times \dots \times \text{Aut}(T_k)) : S_k = (T_1 \times \dots \times T_k) : (\text{Out}(T_1) \times \dots \times \text{Out}(T_k)) : S_k$$

(since $C_G(N) = 1$).

Write $g = (z_1, \dots, z_k)\pi$ where $z_i \in \text{Aut}(T_i)$, $\pi \in S_k$. Let $w = (\delta_1, \dots, \delta_k) \in \Delta_1 \times \dots \times \Delta_k$, $g = (\tau_1, \dots, \tau_k)\pi$. Then

$$\begin{aligned} w^g &= (\delta_1, \dots, \delta_k)^{(z_1, \dots, z_k)\pi} \\ &= (\delta_1^{z_1}, \dots, \delta_k^{z_k})^\pi \\ &= (\delta_1^{z_1\pi^{-1}}, \dots, \delta_k^{z_k\pi^{-1}}) \in \Delta^k. \end{aligned}$$

□

Example 10.0.10. Let $G = (\text{Aut}(T) \times \text{Aut}(T)) : S_2$, $\Omega = \Delta_1 \times \Delta_2 = \Delta^2$, and $G_\omega = (L_1 \times L_2) : S_2$ where $H_i \leq L_i \leq \text{Aut}(T_i)$.

For instance, let $\Delta = \{0, 1, 2, 3, 4\}$, so $\Omega = \Delta^2 = \{(i, j) \mid 0 \leq i, j \leq 4\}$. Fix $\omega = (0, 0)$.

$$G_\omega = (\text{Alt}(1, 2, 3, 4) \times \text{Alt}(1, 2, 3, 4)) : \pi$$

where $\pi : (t_1, t_2) \mapsto (t_2, t_1)$.

Then G_ω partitions $\Omega \setminus \{\omega\} = \Delta^2 \setminus \{(0, 0)\}$ into two orbits:

- $\{(0, i), (i, 0) \mid 1 \leq i \leq 4\}$
- $\{(i, j) \mid 1 \leq i, j \leq 4\}$

We have $(0, 1) \xrightarrow{((12), (12)(34))} (0, 2)$ and $(0, 1) \xrightarrow{\pi} (1, 0)$. Also $(1, 1) \xrightarrow{((k_1, k_2), (k'_1, k'_2))} (1, j)$. Thus $\text{rank}(G) = 3$.

Lemma 10.0.11 (HW). Let $G = (T \times T) : S_2$ acting on $\Omega = \Delta^2$ in product action. Then G is of rank at least 3, and $\text{rank}(G) = 3$ if and only if T is 2-transitive on Δ .

Exercise 10.0.12. Let $G = (T \times T \times T) : S_3$ act on $\Omega = \Delta^3$ in product action. Find a sharp lower bound for $\text{rank}(G)$.

Clue: Let $\Delta = \{0, 1, \dots, n-1\}$, $\Omega = \Delta^k = \{(\delta_1, \dots, \delta_k)\}$, and $\omega = (0, \dots, 0)$. For $w \in \Omega$, define $\text{weight}(w) = \#\{i : \delta_i \neq 0\}$. Then elements of the same weight lie in the same orbit.

Chapter 11

Week 11

$$\text{rank}(G) = 3$$

Theorem 11.0.1. *If $\text{rank}(G) = 3$ and G is quasi-primitive on Ω , then G is HA, AS or $N = T^2$ and G acts on $\Omega = \Delta^2$ in product action with T on Δ being 2-transitive.*

Counterexample: $G = A_5 : S_5 = \text{Aut}(A_5)$ has $\text{rank}(4)$. A_5 acts on A_5 by right multiplication. $S_5 = G_w$ acts on A_5 by conjugation (proved below in Lemma 1). Identify w with 1. Then orbits of G_1 are conjugacy classes of A_5 under S_5 :

$$O_1 = \{1\}, \quad O_2 = \{(ij)(kl)\}, \quad O_3 = \{(ijk)\}, \quad O_4 = \{(ijklm)\}.$$

Lemma 11.0.2. *Let $M \triangleleft G$. G be regular on Ω . Then $\text{rank}(G) \geq \text{number of fusion classes of elements of } M \text{ (orbits under } \text{Aut}(M))$. In particular, if $M = T^k$ with T nonabelian simple, then $\text{rank}(G) \geq 4$. (This implies G is not of type HS, HC, TW.)*

Proof. Identify Ω with M . Let $w \equiv 1$. Then $G = M : G_w$ and G_w acts on $M \equiv \Omega$ by conjugation.

G acts on $\Omega = [G : G_w] = \{G_w g \mid g \in G\}$. For $h \in G_w$: $G_w g \xrightarrow{h} G_w gh = G_w h^{-1} gh = G_w g^h$.

Let $g \in M \Rightarrow g^h \in M$.

Now the action of G_w on M partitions into orbits, i.e., fusion classes of M since $G_w \leq \text{Aut}(M)$. So $\text{rank}(G) \geq \#$ of fusion classes of M .

Suppose $M = T^k$ with T nonabelian simple $\Rightarrow M$ not solvable ($|M| \neq p^a q^b$). Then there exist distinct primes r, s, t dividing $|M|$. Let $x, y, z \in M$ s.t. $|x| = r, |y| = s, |z| = t$. Then $x^{G_w}, y^{G_w}, z^{G_w}$ are three different fusion classes of M . So $\text{rank}(G) \geq 4$. $\square$

Lemma 11.0.3. *Let G be a quasi-primitive group of type SD (simple diagonal). Then $\text{rank}(G) \geq 4$.*

Proof. Let $N = T^k \triangleleft G$ and write $N = T_1 \times \cdots \times T_k$ where $T_1 \simeq \cdots \simeq T_k \simeq T$.

N is regular on Ω ($N = N_w$ is transitive). Identify Ω with $[N : N_w] = \{N_w(t_1, \dots, t_k) \mid t_i \in T\}$, where $N_w(t_1, \dots, t_k) = \{(st_1, \dots, st_k) \mid s \in T\}$ with a representative $(t_k^{-1}t_1, \dots, t_k^{-1}t_{k-1}, 1) \in N$.

Observe $N_w(s_1, \dots, s_{k-1}, 1) = N_w(t_1, \dots, t_{k-1}, 1) \Leftrightarrow (s_1, \dots, s_{k-1}, 1) = (t_1, \dots, t_{k-1}, 1)$.

Now, the stabilizer G_w satisfies $N_w \cdot P \leq G_w \leq \text{Aut}(T) \times S_k$ where $P \leq S_k$ is transitive. (G_w acts by conjugation: $N_w \leq G$, G_w acts on T_i transitively and acts faithfully on N as $C_G(N) \cap N = \{1\} \Rightarrow C_G(N) = \{1\}$. $G_w \leq \text{Aut}(T) \times S_k$ as G_w stabilizes N_w , and $\text{Aut}(T) \times S_k$ is a direct product as $\text{Aut}(T) \times \dots \times \text{Aut}(T)$ becomes $\text{Aut}(T)$, thus S_k acts on $\text{Aut}(T)$ trivially.)

Each element of G_w has the form $\alpha\pi$ with $\alpha \in \text{Aut}(T)$ and $\pi \in S_k$, and an element $(t_1, \dots, t_k) \in N$ is sent by $\alpha\pi$ to $(t_1, \dots, t_k)^{\alpha\pi} = (t_1^\alpha, \dots, t_k^\alpha)^\pi \in N$.

Pick two elements $x := (s_1, 1, \dots, 1)$, $y := (t_1, 1, \dots, 1) \in N \setminus \{(1, \dots, 1)\}$.

Then x, y lie in the same orbit of G_w iff $\exists \alpha\pi \in G_w$ s.t.

$$(s_1, 1, \dots, 1) = (t_1, 1, \dots, 1)^{\alpha\pi} = (t_1^\alpha, 1, \dots, 1)^\pi.$$

1. If $1^\pi = 1$, then $(s_1, 1, \dots, 1) = (t_1^\alpha, 1, \dots, 1)$ so $s_1 = t_1^\alpha$, i.e., s_1, t_1 are conjugated by $\alpha \in \text{Aut}(T)$.
2. If $1^\pi = k$, then $(s_1, 1, \dots, 1) = (1, \dots, 1, t_1^\alpha) = ((t_1^\alpha)^{-1}, \dots, (t_1^\alpha)^{-1}, 1) \in N$. So $s_1 = (t_1^\alpha)^{-1} = (t_1^{-1})^\alpha$ and $k = 2$ (otherwise $(t_1^\alpha)^{-1} = 1$, impossible).
3. If $1^\pi = j \notin \{1, k\}$, say $1^\pi = 2$ wLOG. Then $(s_1, 1, \dots, 1) = (1, t_1^\alpha, 1, \dots, 1) \in N$ so $s_1 = 1$, contradiction.

Therefore $(s_1, 1, \dots, 1)$ and $(t_1, 1, \dots, 1)$ lie in the same orbit of G_w either if $s_1 = t_1^\alpha$ or $k = 2$ and $s_1 = (t_1^{-1})^\alpha$.

Thus $\text{rank}(G) \geq \#$ of orbits of $\text{Aut}(T)$ on $\{t, t^{-1} \mid t \in T\}$. This is ≥ 4 (three distinct primes divide the order of the group; two elements of distinct order are not conjugate nor inverse-conjugate under $\text{Aut}(T)$). $\square$

Definition 11.0.4 (Wreath Product). *The wreath product is defined as*

$$G = T \wr S_k = T^k \rtimes S_k,$$

where the semidirect product is defined by the action

$$(t_1, t_2, \dots, t_k)^\pi = (t_{1\pi^{-1}}, t_{2\pi^{-1}}, \dots, t_{k\pi^{-1}}).$$

Equivalently, the multiplication of two elements is given by

$$((s_1, \dots, s_k)\pi_1)((t_1, t_2, \dots, t_k)\pi_2) = (s_1 t_1^{\pi_1}, s_2 t_2^{\pi_1}, \dots, s_k t_k^{\pi_1})\pi_1 \pi_2.$$

Definition 11.0.5 (Product Action). *Let T be a transitive permutation group on*

$\Delta = \{0, 1, \dots, m-1\}$. For an integer $k \geq 2$, let $P \leq S_k$ be transitive, and let

$$G = T \wr P \leq T \wr S_k \quad \text{and} \quad \Omega = \Delta^k.$$

Then G acts on Ω naturally by

$$(\delta_1, \dots, \delta_k)^{(t_1, \dots, t_k)\pi} = (\delta_1^{t_1\pi^{-1}}, \dots, \delta_k^{t_k\pi^{-1}}).$$

In this way, G is said to be acting on $\Omega = \Delta^k$ in product action, or G is a blow-up of T on Δ .

Fix a point $w_0 = (0, \dots, 0)$. For a point $w = (\delta_1, \dots, \delta_k) \in \Omega$, define the *weight* of w :

$$\text{weight}(w) = |\{i \mid \delta_i \neq 0\}|,$$

the number of nonzero entries of w . The action of each element $(t_1, \dots, t_k)\pi \in G_{w_0}$ preserves the weight of every point of Ω . Thus w_1 is not equivalent to w_2 under G_{w_0} if $\text{weight}(w_1) \neq \text{weight}(w_2)$.

Lemma 11.0.6 (Case: Product Action). *Using the notation defined above ($G = T \wr P \leq T \wr S_k$):*

1. $\text{rank}(G) \geq k + 1$.
2. $\text{rank}(G) = 3$ if and only if $k = 2$ and T is 2-transitive on Δ .

Proof. (1) The weight function takes values in $\{0, 1, \dots, k\}$. So G_{w_0} has at least $k+1$ orbits on Ω .

(2) Assume $\text{rank}(G) = 3$. Then $k = 2$ (since $k \geq 2$), and $\Omega = \Delta \times \Delta$ with $G = (T \times T) \rtimes S_2$. The points of weight equal to 1 form one orbit of G_{w_0} , so $\{(\delta_1, 0) \mid \delta_1 \in \Delta \setminus \{0\}\}$ is an orbit of T_0 , i.e., T is 2-transitive on Δ . The converse is straightforward. $\square$

Chapter 12

Week 12

Let $V = F^d$ ($d > 1$) and let $G = \text{GL}(V)$ act on V with orbits $\{0\}$ and $V \setminus \{0\} := \Omega$.

- G acting on Ω is transitive.

1. If $|F| > 2$, then the action is imprimitive.

Proof: The sets $\{\langle v \rangle \mid v \in V \setminus \{0\}\}$ form a block system $\mathcal{B}$. Note that $|\langle v \rangle| = |F| - 1$.

2. If $|F| = 2$, then the action is 2-transitive.

Proof: Let $(u_1, u_2), (v_1, v_2) \in \Omega \times \Omega$ with $u_1 \neq u_2$ and $v_1 \neq v_2$. Then u_1, u_2 are linearly independent, and so are v_1, v_2 . Extend to bases $\{u_1, \dots, u_d\}$ and $\{v_1, \dots, v_d\}$. There exists $g \in \text{GL}(V)$ such that $u_i^g = v_i$ for all i .

3. If $|F| > 2$, $G = \text{GL}(V)$ acts on $\mathcal{B}$ transitively, with kernel equal to $Z(G)$. Then $G/Z(G) = \text{PGL}(V)$ is 2-transitive on $\mathcal{B}$.
4. $\text{PGL}(V)$ on $\mathcal{B}$ is not 3-transitive if $d > 2$.

Proof: For $d > 2$, choose one set of linearly independent vectors and one set of linearly dependent vectors.

For $d = 2$: Let $\langle u_1 \rangle, \langle u_2 \rangle, \langle u_3 \rangle$ be pairwise different, and $\langle v_1 \rangle, \langle v_2 \rangle, \langle v_3 \rangle$ be pairwise different. Then u_1 and u_2 are linearly independent, and $u_3 = au_1 + bu_2$ with $a, b \in F^\times$. Similarly, v_1 and v_2 are linearly independent, and $v_3 = cv_1 + dv_2$ with $c, d \in F^\times$.

We need to find $g \in \text{GL}(V)$ such that $(u_1, u_2)^g = (v_1, v_2)$ and

$$u_3^g = (au_1 + bu_2)^g = av_1 + bv_2 = v_3 = cv_1 + dv_2.$$

So define g by $u_1 \mapsto (c/a)v_1$ and $u_2 \mapsto (d/b)v_2$, then $u_3 \mapsto v_3$.

Suppose $|F| > 2$. Then $\text{GL}(V)$ acting on $V \setminus \{0\} = \Omega$ is transitive and imprimitive, of rank

$$1 + 1 + |F^\times| - 1 = |F| \quad (F \text{ finite}).$$

Proof: The rank of G^Ω equals the number of G -orbits on $\Omega \times \Omega$. Consider the sets:

$$\{(u, v) \mid \langle u \rangle = \langle v \rangle, u, v \in \Omega\}, \quad \{(u, v) \mid \langle u \rangle \neq \langle v \rangle, u, v \in \Omega\}.$$

If $(u, \lambda u) \mapsto (v, \lambda' v)$ under g , then $u^g = v$ and $\lambda u^g = \lambda v$, so $\lambda = \lambda'$. This partitions into $|F^\times| - 1$ orbits.

Let $N = \mathbb{Z}_p^d$ with p prime and $d \geq 1$. Then $H = \text{Aut}(N) = \text{GL}_d(p)$. Let $G = N \rtimes H$. Let $\Omega = N$ and define an action of $G = \mathbb{Z}_p^d \rtimes \text{GL}_d(p)$ on Ω by

$$w^{xy} = (w + x)^y = w^y + x^y \in N,$$

where the operation in N is addition.

Proposition 12.0.1. *G is an affine 2-transitive permutation group on Ω . Furthermore, if $p = 2$, then G is 3-transitive.*

Proof: Let $(u_1, u_2), (v_1, v_2) \in \Omega \times \Omega$. We need $xy = g \in N \rtimes H$ such that

$$(u_1, u_2)^{xy} = (v_1, v_2),$$

i.e.,

$$((u_1 + x)^y, (u_2 + x)^y) = (v_1, v_2).$$

Equivalently, G is 2-transitive on Ω if and only if G_w is transitive on $\Omega \setminus \{w\}$.

Remark 12.0.2. *It is easy to see that N is a minimal normal subgroup. Every normal subgroup either equals N or is trivial. N is transitive on itself, so G is quasi-primitive $\Rightarrow$ affine.*

Let y map linearly dependent (l.d.) vectors to l.d. vectors and linearly independent (l.i.) vectors to l.i. vectors.

Case 1: Both pairs l.i. There exists $y \in G$ mapping one pair to the other.

Case 2: u_1, u_2 l.i. and v_1, v_2 l.d. Let $x = au_1 + bu_2$.

$$\begin{aligned} u_1 + x &= (1 + a)u_1 + bu_2 \\ u_2 + x &= au_1 + (1 + b)u_2 \quad (\lambda \neq 1). \end{aligned}$$

Then $(1 + a)u_1 + bu_2 = \lambda(au_1 + (1 + b)u_2)$, which gives:

$$1 + a = \lambda a \Rightarrow a = \frac{1}{\lambda - 1}, \quad b = \lambda(1 + b) \Rightarrow b = \frac{\lambda}{1 - \lambda}.$$

Case 3: u_1, u_2 l.d. and v_1, v_2 l.i. Let x be a l.i. combination of u_1 , then $u_1 + x$ and $u_2 + x$ are l.i.

Case 4: u_1, u_2 l.d. and v_1, v_2 l.d. Let $u_1 = \lambda u_2$ and $v_1 = \lambda' v_2$ with $\lambda, \lambda' \neq 1$. Let $x = yu_2$.

$$\frac{x + u_1}{x + u_2} = \frac{\lambda + y}{1 + y} = \lambda' \Rightarrow (x' - 1)y = \lambda - x', \quad y = \frac{\lambda - x'}{x' - 1}.$$

Definition 12.0.3 (k -transitive). *G is said to be k -transitive on Ω if for any k -tuples $(\omega_1, \dots, \omega_k)$ and $(\mu_1, \dots, \mu_k)$ where the ω_i 's are distinct and the μ_i 's are distinct, there*

exists $g \in G$ such that $(\omega_1, \dots, \omega_k)^g = (\mu_1, \dots, \mu_k)$, i.e., $\omega_i^g = \mu_i$ for all i .

Proposition 12.0.4. G is k -transitive if and only if G_ω is $(k-1)$ -transitive on $\Omega \setminus \{\omega\}$ and G is transitive on Ω .

Proof. ($\Rightarrow$) Clear.

($\Leftarrow$) Let $(\omega_1, \dots, \omega_k)$ and $(\mu_1, \dots, \mu_k)$ be k -tuples with distinct entries. We need to find $g \in G$ such that $(\omega_1, \dots, \omega_k)^g = (\mu_1, \dots, \mu_k)$.

Since G is transitive on Ω , there exist $g_1, g_2 \in G$ such that $\omega_1^{g_1} = \omega$ and $\mu_1^{g_2} = \omega$. Then:

$$\begin{aligned} (\omega_1, \dots, \omega_k)^{g_1} &= (\omega, \omega_2^{g_1}, \dots, \omega_k^{g_1}), \\ (\mu_1, \dots, \mu_k)^{g_2} &= (\omega, \mu_2^{g_2}, \dots, \mu_k^{g_2}). \end{aligned}$$

Since G_ω is $(k-1)$ -transitive on $\Omega \setminus \{\omega\}$, there exists $g \in G_\omega$ such that $(\omega_2^{g_1}, \dots, \omega_k^{g_1})^g = (\mu_2^{g_2}, \dots, \mu_k^{g_2})$. Thus $(\omega_1, \dots, \omega_k)^{g_1 g g_2^{-1}} = (\mu_1, \dots, \mu_k)$. $\square$

Exercise 12.0.5. When is a group sharply 2-transitive? (A proper subgroup is no longer 2-transitive.)

Definition 12.0.6 (Sharply k -transitive). G is said to be sharply k -transitive on Ω if for any k -tuples $(\omega_1, \dots, \omega_k)$ and $(\omega'_1, \dots, \omega'_k)$ with distinct entries, there is exactly one element $g \in G$ such that $(\omega_1, \dots, \omega_k)^g = (\omega'_1, \dots, \omega'_k)$.

Example 12.0.7. Let $F = \mathbb{F}_q$ where $q = p^f$. Let $F^+ \cong \mathbb{Z}_p^f$ denote the additive group, and let $F^\times \cong \mathbb{Z}_{p^f-1} = \langle \rho \rangle$ denote the multiplicative group. Then $F^\times = \langle \rho \rangle$ acts on F^+ naturally by field multiplication:

$$v^\rho = v\rho \in F^+.$$

Since ρ is an automorphism of the group F^+ , we have $\rho \in \text{GL}_f(p)$. Define

$$G := F^+ \rtimes \langle \rho \rangle \leq \text{Hol}(F^+) = \mathbb{Z}_{p^f} \rtimes \text{GL}_f(p).$$

Note that $\langle \rho \rangle \cong \mathbb{Z}_{p^f-1}$ is transitive on $F^+ \setminus \{0\}$: for $u, v \in F^+ \setminus \{0\}$, write $u = \rho^i$ and $v = \rho^j$, then $u = v\rho^{i-j}$. So G is 2-transitive on F^+ .

Verification: The number of ordered pairs (ω_1, ω_2) with $\omega_1 \neq \omega_2 \in F^+$ is $p^f(p^f-1) = |G|$, confirming sharp 2-transitivity.

Chapter 13

Week 13

Definition 13.0.1 (Hall subgroup). Let G be a finite group of order $|G| = p_1^{f_1} p_2^{f_2} \dots p_t^{f_t}$ where p_i 's are the prime divisors of $|G|$. Let

$$\pi(G) = \{p_1, \dots, p_t\}.$$

For a subset $\pi \subseteq \pi(G)$, a subgroup $H \leq G$ is called a Hall π -subgroup of G if $\pi(H) = \pi$ and $\gcd(|H|, |G|/|H|) = 1$.

Remark 13.0.2. 1. If $|\pi| = 1$, then a Hall π -subgroup is a Sylow subgroup.

2. Hall subgroups do not necessarily exist.

3. Hall π -subgroups are not necessarily conjugate.

Notation

G_π denotes a Hall π -subgroup (where π is understood as $\pi \cap \pi(G)$). We write G_p if $\pi = \{p\}$, and $\pi' = \pi(G) \setminus \pi$.

Example: $G = A_5$. Then $|G| = 2^2 \cdot 3 \cdot 5$.

- $G_{\{2,3\}} = A_4$
- $G_5 = C_5 = G_{\{2,3\}}'$

Example 13.0.3. 1. $G = A_5$. Then $|G| = 60$. Consider:

$$\pi = \{2, 3\}, \quad \{3, 5\} \text{ (does not exist)}, \quad \{2, 5\} \text{ (does not exist)}.$$

Proof: If H is a Hall $\{2, 5\}$ -subgroup, then $[G : H]$ has size 3. Then $G \hookrightarrow \text{Sym}([G : H]) \simeq S_3$, which is impossible since $G = A_5$ is simple, hence the kernel would be trivial or G , leading to a contradiction. Similarly for $\{3, 5\}$.

2. $G = \text{PSL}_2(11)$ is simple. Then $|G| = 5 \cdot 11 \cdot 12$. This group has subgroups H and K with $H \simeq A_4$ and $K \simeq D_{12}$, which are both Hall $\{2, 3\}$ -subgroups of G .

They are not isomorphic, hence not conjugate.

Theorem 13.0.4 (Hall's Theorem). *Let G be a finite group with $|G| = p_1^{f_1} \cdots p_t^{f_t}$ where $p_1, \dots, p_t$ are distinct primes. Then G is solvable if and only if G has Hall π -subgroups for every subset $\pi \subseteq \pi(G)$.*

Proof. ($\Leftarrow$) We proceed by induction on $|\pi(G)|$.

When $|\pi(G)| = 1$, G is a p -group, hence solvable.

If $|\pi(G)| = 2$, then G is solvable by Burnside's $p^a q^b$ theorem.

Assume $t \geq 3$. Let H_1 be a Hall p_1' -subgroup of G . Then H_1 is solvable by induction since $|\pi(H_1)| = t - 1$.

Let p divide $|H_1|$, i.e., $p = p_j$ for some $j \in \{2, \dots, t\}$. Such a p exists since a minimal normal subgroup of a solvable group is an elementary abelian p -group ($\mathbb{Z}_p^d$ for some p and d), and thus is contained in $O_p(H_1)$.

Then p divides $|H_2|$ or $|H_3|$; say $p \mid |H_2|$. Let P be a Sylow p -subgroup of H_2 . Then P is a Sylow p -subgroup of G , and $O_p(H_1) \leq P^g \leq H_2^g$ for some g by Sylow's theorem. Further, we have $G = H_1 H_2^g$ since

$$|H_1 H_2^g| = \frac{|H_1| \cdot |H_2^g|}{|H_1 \cap H_2^g|} \geq \frac{(p_2^{f_2} p_3^{f_3} \cdots p_t^{f_t})(p_1^{f_1} p_3^{f_3} \cdots p_t^{f_t})}{p_3^{f_3} \cdots p_t^{f_t}} = |G|.$$

Let $N = \langle O_p(H_1)^x \mid x \in G \rangle$, the normal closure of $O_p(H_1)$ in G . Then $N \triangleleft G$ and $N \leq H_2^g$, since for all $x \in G$, we can write $x = x_1 x_2$ for some $x_1 \in H_1$, $x_2 \in H_2^g$, and then

$$O_p(H_1)^x = O_p(H_1)^{x_1 x_2} = O_p(H_1)^{x_2} \leq H_2^g.$$

Thus N is a proper normal subgroup of G .

For a Hall π -subgroup H of G , $H \cap N$ is a Hall π -subgroup of N , and $H/(H \cap N) \simeq HN/N$ is a Hall π -subgroup of G/N .

By induction, N and G/N are both solvable, so G is solvable.

($\Rightarrow$) Let G be solvable. Let M be a minimal normal subgroup of G . Then $M \simeq \mathbb{Z}_p^d$ for some prime p and $d \geq 1$. The quotient $\bar{G} = G/M$ has order properly dividing $|G|$. By induction on the solvable group $\bar{G}$, it has Hall π -subgroups for all $\pi \subseteq \pi(\bar{G})$.

Take a subset $\pi \subseteq \pi(G)$. Let $\bar{H}$ be a Hall π -subgroup of $\bar{G}$.

If $p \in \pi$, then the full preimage of $\bar{H}$ under $G \rightarrow \bar{G}$ is $M \cdot \bar{H}$, which is a Hall π -subgroup of G .

If $p \notin \pi$, then $p \in \pi'$, and we can construct a Hall π -subgroup using the Schur-Zassenhaus theorem applied to the extension $1 \rightarrow M \rightarrow M\bar{H} \rightarrow \bar{H} \rightarrow 1$. $\square$

If $p \notin \pi$, aim to find a subgroup of order $|H|$ of G . Now $(p, |H|) = 1$. Let $X := M \cdot H$ be the full preimage of $\bar{H}$ under $G \rightarrow \bar{G}$. If $X < G$ then by induction X has a Hall π -subgroup H which is isomorphic to $\bar{H}$ and is a Hall π -subgroup of G .

Assume $X = G$. Let $\bar{Q} = O_q(\bar{G}) \neq 1$ for some $q \in \pi(\bar{G})$. Then the full preimage of $\bar{Q}$ under $G \rightarrow \bar{G}$ has the form $M \cdot \bar{Q} = M \rtimes \bar{Q} \triangleleft G$ for some Sylow q -subgroup $Q \simeq \bar{Q}$ of $M \cdot \bar{Q}$ (full preimage of a normal subgroup is a normal subgroup).

(1) Suppose $C_M(Q) \neq 1$.

Then $C_M(Q) \leq Z(M \rtimes Q)$. Since M is abelian, P is a characteristic subgroup of $P \times Q_1$ (Sylow p -subgroup of $P \times Q_1 = Z(M \rtimes Q)$). $Z(M \rtimes Q)$ is a characteristic subgroup of $M \rtimes Q \triangleleft G$, so $P \triangleleft G$. As $P \leq M \triangleleft G$ is minimal, $P = M$ and thus $M \rtimes Q = M \times Q$.

Now Q is characteristic in $M \times Q = M \rtimes Q \triangleleft G$, so $Q \triangleleft G$. Let $\bar{K}$ be a Hall π -subgroup of G/Q . Then the full preimage of $\bar{K}$ under $G \rightarrow G/Q$ is a Hall π -subgroup of G (case $p \in \pi$).

(2) Suppose $C_M(Q) = 1$.

Note that $N_M(Q) = C_M(Q)$ (let $x \in M$ such that $x \in N_M(Q)$, then $xyx^{-1}y^{-1} \in M \cap Q = 1$).

Claim: $N_G(Q) \simeq N_{\bar{G}}(\bar{Q}) = \bar{G}$ (as $\bar{Q} \triangleleft \bar{G}$).

Let $\bar{g} \in \bar{G} = N_{\bar{G}}(\bar{Q})$ and let g be a preimage of $\bar{g}$ in $G \rightarrow \bar{G}$. Then there exists a unique $x \in M$ such that $gx \in N_G(Q)$.

Proof. Since $\bar{g}^{-1}\bar{Q}\bar{g} = \bar{Q}$, we have $g^{-1}(M \rtimes Q)g = M \rtimes Q = M \rtimes g^{-1}Qg$. So $g^{-1}Qg$ is a Sylow q -subgroup of $M \rtimes Q$. By Sylow's theorem, there exists $x \in M$ such that $x^{-1}(g^{-1}Qg)x = Q^{gx} = Q$, i.e., $gx \in N_G(Q)$.

Suppose $gx' \in N_G(Q)$ with $x' \in M$. Then $x^{-1}x' = (gx)^{-1}gx' \in N_G(Q)$. Since $x^{-1}x' \in M$, we have $x^{-1}x' \in N_G(Q) \cap M = N_M(Q) = 1$, so $x = x'$.

Define $\varphi : N_{\bar{G}}(\bar{Q}) \rightarrow N_G(Q)$ by $\bar{g} \mapsto gx$. Suppose $g_1x_1 = g_2x_2$ for some $x_1, x_2 \in M$. Then $g_2^{-1}g_1 = x_2x_1^{-1} \in M$, so $\overline{g_2^{-1}g_1} = \overline{x_2x_1^{-1}} = 1$, hence $\bar{g}_2 = \bar{g}_1$. Thus φ is injective, so $|N_{\bar{G}}(\bar{Q})| \leq |N_G(Q)|$.

Then equality must hold, and we are done (if strict inequality held, then $MN_G(Q)/M \leq G/M = \bar{G}$ while $MN_G(Q)/M = N_G(Q)/(M \cap N_G(Q)) = N_G(Q)/N_M(Q) = N_G(Q)$, contradiction). $\square$

Remark 13.0.5. Moreover, for G solvable, Hall π -subgroups of G are conjugate for any $\pi \subseteq \pi(G)$.

Proof. Let G be solvable. There exists $M < G$ such that $|G/M| = p$ for some prime p (let $N \triangleleft G$ be minimal. Then $\bar{G} = G/N$ has a subgroup $\bar{L}$ such that $|\bar{G}/\bar{L}| = p$ by induction. Let L be the full preimage of $\bar{L}$ under $G \rightarrow \bar{G}$. Then $|G|/|L| = |\bar{G}|/|\bar{L}| = p$).

Let H, K be Hall π -subgroups of G . If $p \notin \pi$, then $H, K \leq M$. By induction, H, K are conjugate in M .

Assume $p \in \pi$. Then $H/(H \cap M) \simeq \mathbb{Z}_p \simeq K/(K \cap M)$ and $H \cap M, K \cap M$ are Hall π -subgroups of M . By induction, $(H \cap M)^x = K \cap M$ for some $x \in M$. Now H^x, K are Hall π -subgroups of G with $K = \langle K \cap M, g \rangle$ for some g with $g^p \in K \cap M$. $\square$

Chapter 14

Week 15

Theorem 14.0.1 (Frattini Argument). *Let G be finite and $N \triangleleft G$. Then for $P \in \text{Syl}_p(N)$ we have $G = N_G(P)N$.*

Proof. By Sylow theorems, N acts transitively on $\text{Syl}_p(N)$ by conjugation. Since $N \triangleleft G$, G induces an action on $\text{Syl}_p(N)$. For any $Q \in \text{Syl}_p(N)$ and $g \in G$, we have $Q^g \in \text{Syl}_p(N)$. As N is transitive on $\text{Syl}_p(N)$, we have $G = NN_G(P)$. $\square$

Remark 14.0.2. 1. *Let G be a p -group, then $Z(G) \neq 1$ and $Z(G/Z(G)) \neq 1$ (if G is non-abelian). Let K be the full preimage of $Z(G/Z(G))$ under $G \rightarrow G/Z(G)$. Then $Z(G) \triangleleft K \triangleleft G$. Continuing in this way we get a normal series.*

2. *Let $G = P \times Q$, where P is a p -group, Q is a q -group, p, q distinct primes. Then $Z(G) \neq 1$ and $G/Z(G) \cong \bar{P} \times \bar{Q}$.*

Definition 14.0.3. *A group G is called nilpotent if it has a chain of subgroups:*

$$G = K_1 > K_2 > K_3 > \cdots > K_{s+1} = 1$$

such that $K_i \triangleleft G$ and $K_i/K_{i+1} \leq Z(G/K_{i+1})$. Such a series is called a central series.

Example 14.0.4 (Non-nilpotent groups). 1. $A_4 = C_2^2 \rtimes C_3$ has only one proper nontrivial normal subgroup C_2^2 . The chain $A_4 > C_2^2 > 1$ fails since $C_2^2/1 \not\leq Z(A_4)$.

2. S_4 has normal subgroups A_4, C_2^2 . The chain $S_4 > A_4 > C_2^2 > 1$ is not possible since $C_2^2 \neq Z(S_4)$.

3. D_{2p} for $p \geq 3$ prime. It has only one proper nontrivial normal subgroup $\langle a \rangle$ where $D_{2p} = \langle a, b \mid a^p = b^2 = 1, a^b = a^{-1} \rangle$. The chain $D_{2p} > \langle a \rangle > 1$ fails since $\langle a \rangle \neq Z(D_{2p})$.

Theorem 14.0.5 (Equivalent definitions of nilpotent groups). *The following are equivalent:*

1. G is nilpotent.
2. If $H < G$, then $N_G(H) > H$.
3. Each maximal subgroup M of G is normal and is of prime index (i.e., $G/M \cong \mathbb{Z}_p$ for some p).
4. G is a direct product of its Sylow subgroups, i.e., each Sylow subgroup is normal.

Example 14.0.6. 1. Abelian groups are nilpotent groups.

2. Subgroups and factor groups of nilpotent groups are nilpotent.
3. $G = A_4 = C_2^2 \rtimes C_3$. The subgroup C_3 is maximal but not normal, hence not of prime index.

Proof of (1) $\Rightarrow$ (2). G has a chain of subgroups

$$G = K_1 > K_2 > \cdots > K_{s+1} = 1.$$

Let $H \leq G$. Then there exists i such that $K_{i+1} \leq H$ and $K_i \not\leq H$.

Then H/K_{i+1} and $Z(G/K_{i+1})$ centralize each other. As $K_i/K_{i+1} \leq Z(G/K_{i+1})$, for any $h \in H$, $y \in K_i$, we have $hyh^{-1}y^{-1} \in K_{i+1} \subset H$.

Hence $yh^{-1}y^{-1} \in h^{-1}H = H$, i.e., $y \in N_G(H)$. Thus $K_i \leq N_G(H)$.

Therefore $N_G(H) \geq HK_i > H$ as required. $\square$

Proof of (2) $\Rightarrow$ (3). $\square$

Let $M \triangleleft G$ be maximal. Then $M \triangleleft G$. Then G/M does not have a nonidentity proper subgroup as M is maximal, so $|G/M|$ is prime as in (3).

(3) $\Rightarrow$ (4): Let $P \in \text{Syl}_p(G)$. Then $P < G$ and $P < N_G(P) \leq G$. Suppose $N_G(P) \neq G$. Then $\exists M$ s.t. $N_G(P) < M < G$. Then $M \triangleleft G$ and by Frattini argument, $G = N_G(P)M = M$, contradiction. Hence $P \triangleleft N_G(P) = G$ as in (4).

(4) $\Rightarrow$ (1): Let $|G| = p_1^{e_1} p_2^{e_2} \cdots p_t^{e_t}$ with $p_1, \dots, p_t$ distinct primes. Let $P_i \in \text{Syl}_{p_i}(G)$. Then $P_i \triangleleft G$ and $G = P_1 \times \cdots \times P_t$. Let $L_1 = Z(G)$. Then $L_1 \neq 1$ and $\bar{G} = G/L_1$ is again a direct sum of its Sylow subgroups. Let $\bar{L}_2 = Z(\bar{G})$ and let L_2 be the full preimage of $\bar{L}_2$ under $G \rightarrow \bar{G}$. Then $L_2 \triangleleft G$ as $\bar{L}_2 \triangleleft \bar{G}$ and $L_2/L_1 = Z(G/L_1)$. Let $\bar{L}_3 = Z(G/L_2)$ and let L_3 be the full preimage of $\bar{L}_3$ under $G \rightarrow G/L_2$. Then $L_3 \triangleleft G$ and $L_3/L_2 = Z(G/L_2)$. Repeating this process produces a chain of subgroups

$$1 < L_1 < L_2 < L_3 < \cdots < L_s = G$$

such that $L_i \triangleleft G$ and $L_i/L_{i-1} = Z(G/L_{i-1})$ as in (1). $\square$

Frattini subgroup and nilpotent groups

Definition 14.0.7 (Frattini subgroup). *Let G be a finite group. Then G has maximal subgroups, say $\text{Max}(G) = \{M < G\}$. The intersection of all maximal subgroups is called the Frattini subgroup of G , denoted by $\Phi(G)$.*

- Corollary 14.0.8.**
1. $\Phi(G)$ is nilpotent.
 2. G nilpotent $\Leftrightarrow G/\Phi(G)$ nilpotent.
 3. (Wielandt) G nilpotent $\Leftrightarrow G' \leq \Phi(G)$.
 4. If G is a p -group, then $G/\Phi(G)$ is elementary abelian.

Remark 14.0.9 (Observation).

1. $\Phi(G)$ is a characteristic subgroup of G .
2. If $N \triangleleft G$, then $\Phi(N) \triangleleft G$ (by (1)).

Definition 14.0.10 (Non-generator). *An element x of G is called a non-generator if $\langle x, S \rangle = G$ implies $\langle S \rangle = G$ for any $S \subseteq G$.*

- Example 14.0.11.**
1. $G = \langle a \rangle = C_{p^2}$. Then each element of $\langle a^p \rangle$ is a non-generator.
 2. Q_8 has non-generators of order 1 or 2.
 3. D_{2n} .
 4. A nonabelian simple group has no nonidentity non-generator element. In fact, $\forall a, \exists b$ s.t. $\langle a, b \rangle = G$.

Lemma 14.0.12 (Characterization of $\Phi(G)$). $\Phi(G)$ consists of all non-generators of G .

Proof. Let $x \in \Phi(G)$. Suppose $\langle x, S \rangle = G$ but $\langle S \rangle \neq G$. Then $\exists M < G$ s.t. $\langle S \rangle \leq M < G$. Then $x \in \Phi(G) \leq M$, so $G = \langle x, S \rangle \subseteq M$, not possible. Hence x is a non-generator.

Let $x \in G$ be a non-generator. Suppose $M < G$ and $x \notin M$. Then $\langle x, M \rangle = G$, not possible as $M < G$. So $x \in M$. Since this holds for all maximal M , we have $x \in \Phi(G)$. $\square$

Theorem 14.0.13 (Gaschütz). *Let $N \triangleleft G$ s.t. $N \leq \Phi(G)$. Then G/N is nilpotent $\Leftrightarrow G$ is nilpotent.*

Proof. It suffices to prove $\Rightarrow$. Let $P \in \text{Syl}_p(G)$. We will show $P \triangleleft G$. Then $PN/N \in \text{Syl}_p(G/N)$ and $PN/N \triangleleft G/N$. Hence $PN \triangleleft G$. By Frattini argument,

$$G = PNN_G(P) \xrightarrow{N \leq \Phi(G)} PN_G(P) = N_G(P),$$

so $P \triangleleft G$, and thus G is nilpotent. $\square$

Proof of Corollary. 1. It suffices to prove any Sylow p -subgroup P of $\Phi(G)$ is normal in $\Phi(G)$. By Frattini argument, $G = \Phi(G)N_G(P)$. If $N_G(P) \neq G$, $\exists M$ s.t. $N_G(P) \leq M < G$. But $\Phi(G) \leq M$, so $G \leq M$, contradiction. Hence $P \triangleleft G$.

2. Take $N = \Phi(G)$ in the theorem.

3. Suppose G is nilpotent. Then for any $M < G$, $M \triangleleft G$ and $G/M \simeq C_p$ is abelian. Thus $G' \leq M$ and $G' \leq \Phi(G)$.

4. By (3), $G/\Phi(G)$ is abelian. Moreover, $x^p \in \Phi(G)$ for all $x \in G$. Indeed, let $M < G$. If $x \in M$, we are done. If $x \notin M$,

$\square$

Since $G/M \simeq C_p$, we have $x^p \in M$. Then $\forall \bar{g} \in G/\Phi(G)$, $\bar{g}^p = 1$. Hence $G/\Phi(G)$ is elementary abelian.

Example 14.0.14. Let $G = C_p^2 \rtimes C_q = \langle a \rangle \rtimes \langle b \rangle$, where $p \neq q$ are primes. Then $a^p b = b a^p \Leftrightarrow ab = ba \Leftrightarrow G/\langle a^p \rangle$ is abelian. In particular, G abelian $\Leftrightarrow G/\langle a^p \rangle \simeq C_p \rtimes C_q$ abelian.

Definition 14.0.15 (Fitting subgroup). The product of all normal nilpotent subgroups of G is called the Fitting subgroup of G , denoted by $F(G)$. Note that $\Phi(G) \leq F(G)$.

Theorem 14.0.16. Let G be solvable, and $F = F(G)$. Let $|G| = p_1^{e_1} p_2^{e_2} \cdots p_t^{e_t}$. Then:

1. $F = O_{p_1}(G) \times \cdots \times O_{p_t}(G)$ (characterization).
2. $C_G(F) \leq F$, i.e., F is self-centralizing. Consequently, $G/C_G(F) \leq \text{Aut}(F)$.

Remark 14.0.17. Recall $O_p(G)$. Let $p \mid |G|$. Then

$$O_p(G) = \begin{cases} \text{the largest normal } p\text{-subgroup of } G, \\ \text{the product of all normal } p\text{-subgroups of } G, \\ \bigcap_{P \in \text{Syl}_p(G)} P. \end{cases}$$

1. $O_p(G) \text{ char } G$ ($O_p(G)$ is a characteristic subgroup) and $\bar{G} := G/O_p(G)$ satisfies $O_p(\bar{G}) = 1$.
2. If $q \neq p$, then $\langle O_p(G), O_q(G) \rangle = O_p(G) \times O_q(G)$.

Proof. (1) Let N be a normal nilpotent subgroup of G . Then $N = Q_1 \times Q_2 \times \cdots \times Q_t$ where $Q_i \in \text{Syl}_{p_i}(N)$. Now $Q_i \text{ char } N \triangleleft G$, so $Q_i \triangleleft G$ and thus $Q_i \leq O_{p_i}(G)$. Hence $N \leq O_{p_1}(G) \times \cdots \times O_{p_t}(G)$. Since $O_{p_1}(G) \times \cdots \times O_{p_t}(G)$ is a normal nilpotent subgroup of G , we have $F = O_{p_1}(G) \times \cdots \times O_{p_t}(G)$.

(2) Let $C = C_G(F)$. Then $C \cap F \leq Z(F)$. Since $F \text{ char } G$, so are C , CF and $C \cap F$. Let $W = C \cap F$. Let $\bar{G} := G/W$. Suppose $\bar{C} \neq 1$. Then there exists p such that $O_p(\bar{C}) \neq 1$, and $O_p(\bar{C}) \text{ char } \bar{C} \text{ char } \bar{G}$. Then the full preimage of $O_p(\bar{C})$, say $Y := W \cdot O_p(\bar{C}) \leq C$, is characteristic in G . Let $P \in \text{Syl}_p(Y)$. Then $W_p \leq P \leq C$. By definition of W and C , P centralizes W_p , so $Y = W_p \times P$. Then $P \text{ char } Y \text{ char } G$ and $P \triangleleft G$, so that $W_p < P \leq F$, a contradiction. Thus $O_p(\bar{C}) = 1$ and $C \leq Z(F) \leq F$. $\square$

Example 14.0.18. *Let G be such that each Sylow subgroup is cyclic. Then G is solvable. Furthermore, G is metacyclic, i.e., there exists $C \triangleleft G$ such that C is cyclic and G/C is cyclic.*

Proof. Let $F = F(G)$. Then F is cyclic and $G/C_G(F) \lesssim \text{Aut}(F)$ is abelian. Since F is abelian, $C_G(F) = F$ and G/F is abelian. Since Sylow subgroups of G/F are cyclic, G/F is cyclic. Hence G is solvable. $\square$

Exercise 14.0.19. 1. *Let G be as in the example above.*

2. *Let G be solvable and $|G| = 4m$ where m is odd and square-free. Characterize G .*